HPSC & HSSC SCREENING EXAM COMPREHENSIVE DOSSIER

# HARYANA GENERAL KNOWLEDGE

Master Preparation Keynote for Food Safety Officer (FSO), HCS & Screening Tests

### I. HISTORY & ARCHAEOLOGY

Vedic Roots, Rakhigarhi, 3 Battles of Panipat, 1857 Heroes & Chhotu Ram

### II. GEOGRAPHY & HYDROLOGY

Shivalik to Aravalli, Yamuna-Ghaggar, Canals, Soils, Forests & Sanctuaries

### III. AGRICULTURE & FOOD ECOSYSTEM

FSO Special: Murrah Buffalo, NDRI, NIFTEM, Mega Food Parks & Crop Belts

### IV. POLITY, ECONOMY & DISTRICTS

90 Vidhan Sabha, Census 2011, 22 Districts Dossier & Top 30 Exam Traps

*Based on Comprehensive Cross-Examination Analysis of 10+ Years HPSC / HSSC Screening Question Papers*

# Weightage & Question Distribution in Haryana Screening Tests

*Key Exam Takeaway: Haryana GK carries 20-25% decisive weightage in HPSC FSO and ADO screening tests.*

| Exam Category | Conducting Body | Total Questions | Haryana GK Weightage | Core High-Frequency Focus Areas |
|---|---|---|---|---|
| **Food Safety Officer (FSO)** | HPSC | 100 MCQs | 20-25 Questions | Agriculture, Murrah Buffalo, NDRI/NIFTEM, Food Laws, Soil & Geography |
| **HCS (Executive Branch)** | HPSC | 100 MCQs (Paper 1) | 15-20 Questions | Ancient History, Harappan Sites, 1857 Revolt, Vidhan Sabha Polity, Schemes |
| **Agriculture Development Officer** | HPSC | 100 MCQs | 20 Questions | Soils, Canal Irrigation (WYC/Bhakra), Crop Seasons, Agro-Climatic Zones |
| **Assistant Professor (College)** | HPSC | 100 MCQs | 15-20 Questions | History, Folk Literature (Saang), Census 2011, Demographics, Personalities |
| **HSSC CET (Group C / D)** | HSSC | 100 MCQs | 25 Questions (Fixed 25%) | District Profiles, Folk Dances, Rivers, Sports Stars, Wildlife Sanctuaries |

### FSO SCREENING STRATEGIC MANDATE

In the Food Safety Officer (FSO) screening test, subject knowledge (Food Tech, Microbiology, Chemistry, FSS Act 2006) accounts for ~80% of technical marks, while Haryana GK forms the mandatory non-technical section (20%). Candidate qualification cutoff hinges on this section because technical scores are closely grouped. High-yield topics include: Dairy Research Infrastructure (NDRI, CIRB, LUVAS), Basmati Rice & Mustard Zones, Mega Food Parks, Soil Salinity Reclamation (CSSRI), and Government Agro-Nutrition Initiatives.

Analysis of HPSC FSO 2023, HPSC ADO 2022-23, HPSC HCS Pre 2021-2024, and HSSC CET Group C/D Exam Patterns.

# The Political Journey to Statehood: 1858 to 1966

*Key Exam Takeaway: Haryana was tagged to Punjab in 1858 as British collective punishment for active participation in the 1857 revolt.*

## PHASE I: PRE-INDEPENDENCE DEMANDS

• **Colonial Annexation (1858):** After the 1857 First War of Independence, the British detached Haryana from North-Western Provinces and merged it with Punjab to penalize the rebellious peasantry.

• **First Formal Demand (1926):** All India Muslim League Delhi session, Pirzada Muhammad Hussain demanded separation of Delhi and Haryana territory from Punjab.

• **Deshbandhu Gupta's Advocacy (1932):** Formulated a clear plea for 'Greater Delhi' integrating Ambala Division, Meerut, and Agra into a culturally unified linguistic zone.

• **Dinbandhu Sir Chhotu Ram:** Stood for peasant rights through the Unionist Party (1923), safeguarding rural agrarian interests under the Punjab Legislative Council.

## PHASE II: POST-INDEPENDENCE FORMULAS

• **Sachar Formula (1 Oct 1949):** Proposed by CM Bhim Sen Sachar. Divided Punjab into 'Punjabi Zone' (Gurmukhi script) and 'Hindi Zone' (Devanagari script). Rejected due to language friction.

• **Fazal Ali Commission (SRC 1953-55):** States Reorganisation Commission rejected separate Haryana, citing administrative synergy, causing massive discontent among Hindi speakers.

• **Regional Formula (1956):** Established Hindi and Punjabi Regional Committees in Punjab Assembly. Failed to resolve deep developmental and cultural disparities.

• **Sant Fateh Singh's Agitation (1965):** Threat of self-immolation for Punjabi Suba forced Central Government to form Parliamentary Committee headed by Lok Sabha Speaker Hukam Singh.

# Shah Commission & 18th Amendment: Birth of 17th State

*Key Exam Takeaway: Haryana was created under the Punjab Reorganisation Act 1966 via the 18th Constitutional Amendment.*

## SHAH COMMISSION KEY PARAMETERS

• **Constitution Date:** 23 April 1966 by the Government of India under the chairmanship of Justice J.C. Shah (Supreme Court Judge).

• **Commission Members:** Justice J.C. Shah (Chairman), S. Dutt (Member), and M.M. Philip (Member).

• **Mandate:** Demarcate the Hindi-speaking areas of the composite State of Punjab to carve out a new State.

• **Report Submission:** 31 May 1966. Recommended creation of Haryana with Ambala, Gurgaon, Hissar, Rohtak, Mahendragarh, Jind, and Narwana tehsils.

• **Kharar & Chandigarh Controversy:** Commission recommended Kharar tehsil (including Chandigarh) to Haryana by 2:1 majority (S. Dutt dissented). Center later made Chandigarh a Joint Capital Union Territory.

## CONSTITUTIONAL & LEGISLATIVE FRAMEWORK

• **18th Constitutional Amendment Act 1966:** Clarified parliamentary power under Article 3 to form a new State or Union Territory by uniting parts of existing states.

• **Punjab Reorganisation Act 1966:** Passed by Parliament in September 1966; received Presidential assent on 18 September 1966.

• **Official Formation:** 1 November 1966 as the 17th State of the Republic of India.

• **Area Ceded:** Approximately 35.18% of the composite Punjab area (44,212 sq km) was transferred to Haryana.

• **Shared Institutions:** Common High Court (Punjab & Haryana High Court at Chandigarh), Joint Capital (Chandigarh), and shared Bhakra Beas Management Board (BBMB).

# District Genesis: From Original 7 ('GHAR JK') to 22 Districts

*Key Exam Takeaway: Haryana began on Nov 1, 1966 with 7 districts; Charkhi Dadri was carved out as the 22nd district in Dec 2016.*

| Chronological Phase | Districts Created | Carved Out From | Formation Date | Exam Significance |
|---|---|---|---|---|
| **Original 7 (At Birth)** | Gurgaon, Hissar, Ambala, Rohtak, Jind, Karnal, Mahendragarh | Punjab Reorganisation Act | 1 Nov 1966 | Acronym 'GHAR JK'. Largest at birth: Hissar; Smallest: Jind. |
| **Phase 2 (1972-73)** | Bhiwani & Sonipat (1972)<br>Kurukshetra (1973) | Hissar/Rohtak<br>Karnal | 22 Dec 1972<br>23 Jan 1973 | Bhiwani created by Bansi Lal; Kurukshetra named after King Kuru. |
| **Phase 3 (1975-79)** | Sirsa (1975)<br>Faridabad (1979) | Hissar<br>Gurgaon | 1 Sep 1975<br>2 Aug 1979 | Sirsa became largest district after Bhiwani split. Faridabad industrial hub. |
| **Phase 4 (1989 Quad)** | Kaithal, Panipat, Rewari, Yamunanagar | Karnal, Rohtak, Gurgaon, Ambala | 1 Nov 1989 | Created simultaneously on Haryana Day. Panipat merged back & re-created 1992! |
| **Phase 5 (1995-97)** | Panchkula (1995)<br>Fatehabad & Jhajjar (1997) | Ambala<br>Hissar & Rohtak | 15 Aug 1995<br>15 Jul 1997 | Panchkula planned satellite; Fatehabad historical city of Firoz Shah. |
| **Phase 6 (2005-16)** | Nuh/Mewat (2005), Palwal (2008), Charkhi Dadri (2016) | Gurgaon/Faridabad<br>Bhiwani | 4 Apr 2005<br>15 Aug 2008<br>1 Dec 2016 | Nuh created as Satyamevpuram. Charkhi Dadri is the 22nd & newest district. |

Department of Revenue & Disaster Management, Haryana; State Statistical Abstract 2023-24.

# Official State Symbols & Emblems of Haryana

*Key Exam Takeaway: Black Francolin (Kala Teetar) is State Bird; Peepal is State Tree; Lotus is State Flower.*

## STATE BIRD: BLACK FRANCOLIN

- **Scientific Name:** Francolinus francolinus.
- **Local Name:** Kala Teetar (काला तीतर).
- **Habitat:** Shrublands, cultivated fields of Sirsa, Hisar, Bhiwani, and Kurukshetra.
- **Ecological Role:** Natural pest predator in wheat and mustard fields.
- **HPSC Trap:** Do not confuse with Great Indian Bustard (Rajasthan) or Monal (Uttarakhand).

## STATE ANIMAL: BLACKBUCK

- **Scientific Name:** Antilope cervicapra.
- **Local Name:** Krishna Mriga / Kala Hiran (काला हिरण).
- **Primary Range:** Abubshahar Wildlife Sanctuary (Sirsa), Bishnoi settlements in Fatehabad and Hisar.
- **Cultural Protector:** Bishnoi community upholds absolute sacred protection.
- **IUCN Status:** Least Concern (Schedule I of Wildlife Protection Act 1972).

## STATE TREE: PEEPAL

- **Scientific Name:** Ficus religiosa.
- **Local Name:** Peepal (पीपल) / Bodhi Vriksha.
- **Sacred Significance:** Mentioned in Rigveda and Bhagavad Gita (Lord Krishna calls himself Ashvattha among trees).
- **Medicinal & Oxygen Value:** Continuous oxygen producer; rich in flavonoids and tannins used in traditional medicine.
- **HPSC Trap:** State Tree is Peepal, NOT Khejri (which is Rajasthan's State Tree).

## FLOWER, SEAL & LANGUAGE

- **State Flower:** Lotus (Nelumbo nucifera) / Kamal. Sacred aquatic herb prevalent in village ponds and lakes.
- **Official State Seal:** Circular seal featuring Lion Capital of Ashoka with a rising sun emerging behind lotus blooms, flanked by ears of wheat.
- **Official Language:** Hindi (Devanagari script). Punjabi was declared 2nd official language in 2010 (under Bhupinder Singh Hooda; earlier Bansi Lal made Tamil 2nd language in 1969 to counter Punjabi).
- **State Sport:** Wrestling (Kushti) - Traditional soil akhadas producing 60%+ of national Olympic wrestling medals.

# Geographical Coordinates, Territorial Extent & Border States

*Key Exam Takeaway: Haryana covers 44,212 sq km (1.34% of India) and is landlocked by 4 States & 2 UTs.*

## COORDINATES & PHYSICAL EXTENT

- **Latitudinal Extent:** 27°39' N to 30°55' N latitude (Span: ~3°16').
- **Longitudinal Extent:** 74°27' E to 77°36' E longitude (Span: ~3°09').
- **Total Geographical Area:** 44,212 sq km (17,070 sq miles), accounting for 1.34% of India's total land area.
- **National Area Rank:** 20th among Indian States (21st including UT of Ladakh/J&K reorganisation).
- **Elevation Profile:** Ranging from 200 m (656 ft) in southwestern depressions to 1,514 m in the northern Morni Hills.

## BOUNDARIES & BORDER DYNAMICS

- **Bordering States (4 States):** Punjab (North-West), Himachal Pradesh (North), Uttar Pradesh (East), and Rajasthan (South & South-West).
- **Bordering Union Territories (2 UTs):** New Delhi (surrounded on three sides: North, West, South) and Chandigarh (Joint Capital in North).
- **Longest Inter-State Border:** Rajasthan shares the longest boundary (~1,262 km) with 7 Haryana districts (Sirsa, Fatehabad, Hisar, Bhiwani, Mahendragarh, Rewari, Nuh).
- **Shortest Inter-State Border:** Himachal Pradesh shares ~11 km border with Panchkula, Ambala, and Yamunanagar.
- **Landlocked Nature:** Completely enclosed with zero international border and no coastal outlet.
- **Districts with No Inter-State Boundary:** Rohtak and Charkhi Dadri (touch only internal Haryana districts).

# Physiography of Haryana: Dr. Jasbir Singh's 8-Fold Classification

*Key Exam Takeaway: In 'Agricultural Geography of Haryana', Dr. Jasbir Singh classified the terrain into 8 distinct relief units.*

| Physiographic Unit | Geographical Coverage | Topographical Nature | Key Soil / Ecological Feature |
|---|---|---|---|
| **1. Shivalik Hills** | Panchkula, Northern Ambala & Yamunanagar | Tertiary folded hills; altitude 400m - 1514m | Subject to severe gully erosion; Pinjore gravel zone |
| **2. Piedmont Plain (Ghar/Kandi)** | Foothills of Shivaliks (Panchkula, Ambala) | Piedmont alluvial fans formed by seasonal torrents | Coarse sand, boulders; dissected by 'Chos' (torrents) |
| **3. Alluvial Plain (Bangar)** | Central Haryana (Kaithal, Karnal, Jind, Rohtak) | Flat, older alluvium plain; elevation 220m - 250m | Highly fertile loam; heart of Green Revolution |
| **4. Flood Plains (Khadar/Naili/Bet)** | Yamuna, Ghaggar, and Markanda flood belts | Low-lying newer alluvium formed by seasonal overflows | Khadar (Yamuna), Naili (Ghaggar), Bet (Markanda) |
| **5. Sandy Plains & Sand Dunes** | Sirsa, Fatehabad, Hisar, Bhiwani, Mahendragarh | Aeolian undulating plains with shifting sand dunes (Tibas) | Semi-arid; high wind erosion; calcareous dry soils |
| **6. Aravalli Outliers** | Gurugram, Mewat (Nuh), Rewari, Mahendragarh | Denuded Precambrian residual metamorphic ridges | Quartzite, slate, granite outcrops; rocky thin soils |
| **7. Drifting Sand Zone (Taal)** | Southwestern desert transition belt (Bhiwani/Sirsa) | Shallow depressions between sand dunes | Saline water accumulation during episodic rain |
| **8. Marshy / Tarai Tract** | Southern Sirsa (Sanisa depression / seasonal marshes) | Low-lying water accumulating basins | Heavy clayey soils; prone to temporary waterlogging |

# The Shivalik Range, Morni Hills & Karoh Peak Profile

*Key Exam Takeaway: Karoh Peak (1,514m) in Morni Hills, Panchkula is the highest elevation point of Haryana.*

## SHIVALIK RELIEF & MORNI HILLS

• **Geological Origin:** Outer-most Himalayan foothills formed during late Cenozoic era by fluvial deposits folded during Himalayan orogeny.

• **Hypsometric Peak:** Karoh Peak (करोह चोटी) situated in Morni Hills, Panchkula district at an elevation of 1,514 metres (4,967 ft) - widely asked in HPSC/HSSC.

• **Morni Hills Ecosystem:** The only hill station of Haryana, home to Tikkar Taal (twin lakes: Chhota Tikkar and Bada Tikkar separated by hillock).

• **Drainage Origin:** Cradle of seasonal foothill streams: Tangri, Ghaggar tributaries, and Markanda torrents.

• **Vegetation:** Sub-tropical chir pine (Pinus roxburghii), mixed deciduous forests, and rich wild medicinal flora.

## GHAR, KANDI & THE CHERRAPUNJI OF HARYANA

• **Piedmont Plain (Ghar & Kandi):** Narrow alluvial strip (10-15 km wide) lying immediately south of Shivaliks across Panchkula, Ambala, and Yamunanagar.

• **Local Terminology Trap:** Known locally as 'Ghar' (घर) in Kalka/Raipur Rani and 'Kandi' (कांडी) in hilly margins.

• **Torrents ('Chos'):** Cut deep ravines and seasonal dry beds called Chos that cause suddenflash floods during monsoon.

• **Chhachhrauli Phenomenon:** Located in Yamunanagar district; receives highest average annual rainfall in the state (>1100 mm), famously dubbed 'The Cherrapunji of Haryana'.

• **Geological Assets:** Boulder gravel, sandstone beds, and lime mortar kankar deposits extensively quarried for regional construction.

# Aravalli Outliers, Residual Metamorphic Ridges & Dhosi Hill

*Key Exam Takeaway: Dhosi Hill (652m) in Mahendragarh is the highest peak of Aravallis in Haryana and the hermitage of Rishi Chyavana.*

## DHOSI HILL & ARAVALLI RIDGES

• **Precambrian Metamorphic Antiquity:** One of the oldest fold mountain chains in the world (Delhi Supergroup: Alwar and Ajabgarh series).

• **Geographical Spread:** Discontinuous relict hills entering through Rajasthan into Mahendragarh, Rewari, Gurugram, Nuh (Mewat), and Faridabad.

• **Highest Elevation Peak:** Dhosi Hill (धोसी की पहाड़ी) at 652 metres (2,139 ft) situated at the border of Mahendragarh (near Narnaul) and Jhunjhunu (Rajasthan).

• **Extinct Volcanic Crater:** An isolated volcanic plug of late Proterozoic era with a crater-like depression at the summit.

• **Ecological Shield:** Acts as a critical green natural barrier preventing eastward expansion of the Thar Desert into Indo-Gangetic plains.

## HISTORICAL, MEDICINAL & ECOLOGICAL ROLE

• **Vedic Hermitage of Chyavana Rishi:** Ancient seat where sage Chyavana (son of sage Bhrigu) formulated the legendary Ayurvedic preparation 'Chyawanprash' using wild Amla and herbs.

• **Surajkund Sun Temple & Amphitheatre:** Built by Tomar King Anangpal I in 10th century CE in the Aravalli foothills of Faridabad.

• **Kotla & Khalilpur Depressions:** Natural drainage basins at the foot of Mewat Aravalli hills collecting seasonal runoff.

• **Mining & Conservation Ban:** Supreme Court of India banned commercial mining in Aravalli ridges across Faridabad and Gurugram to restore groundwater aquifers.

• **Aravalli Green Wall Project:** Ambitious national 1,400 km long and 5 km wide afforestation green belt spanning Haryana, Rajasthan, Gujarat, and Delhi.

# Soil Classification, Texture Profile & Agro-Ecological Zones

*Key Exam Takeaway: Soils vary from light sandy loams in the south-west to fertile alluvial loams in central plains and rocky loams in Shivaliks.*

## SOIL TEXTURAL CLASSES

• **Very Light Soils (Sandy):** Found in Sirsa, Fatehabad, Bhiwani, and Mahendragarh. High permeability, deficient in nitrogen and organic matter, prone to wind erosion.

• **Light Soils (Sandy Loam):** Spans dry parts of Hisar, Rewari, and Jhajjar. Highly responsive to tubewell/canal irrigation and phosphatic fertilizers.

• **Medium Soils (Loam & Silt Loam):** Dominant in Kaithal, Jind, Sonipat, Panipat, and Rohtak. The backbone of wheat, basmati paddy, and sugarcane cropping.

• **Moderately Heavy & Heavy Soils (Clayey):** Trapped along Ghaggar basin (Naili tract) and parts of Yamunanagar and Karnal; rich in nutrients but slow drainage.

## AGRO-ECOLOGICAL DISTRIBUTION

• **Agro-Climatic Zone I (Eastern / Wet Zone):** Panchkula, Ambala, Yamunanagar, Kurukshetra, Karnal, Kaithal, Sonipat, Panipat. Sub-humid to semi-arid, high irrigation, paddy-wheat rotation.

• **Agro-Climatic Zone II (Western / Dry Zone):** Sirsa, Fatehabad, Hisar, Bhiwani, Charkhi Dadri, Mahendragarh, Rewari, Jhajjar. Arid to semi-arid, mustard-pearl millet-cotton cultivation.

• **Organic Carbon Status:** Generally low to medium (<0.5%) due to intensive double-cropping and crop residue burning.

• **Micronutrient Deficiencies:** Zinc (Zn) and Iron (Fe) widespread across central plains; Sulphur (S) deficit pronounced in oilseed mustard belts.

# Soil Salinity, Alkalinity (Kallar / Rehi) & CSSRI Reclamation

*Key Exam Takeaway: CSSRI Karnal pioneered gypsum application and sub-surface drainage (SSD) to reclaim sodic soils.*

| Diagnostic Property | Saline Soils (Solonchak) | Alkali / Sodic Soils (Kallar/Rehi) | CSSRI Reclamation Technology |
|---|---|---|---|
| **pH Range** | < 8.5 (Neutral to mildly alkaline) | > 8.5 (Highly alkaline, up to 10.5) | Gypsum (CaSO4.2H2O) requirement based on Soil Testing |
| **Electrical Conductivity (EC)** | > 4.0 dS/m (High soluble salts) | < 4.0 dS/m (Low neutral soluble salts) | Leaching with good quality canal water through deep furrows |
| **Exchangeable Sodium (ESP)** | < 15% (Low sodium saturation) | > 15% (Excessive exchangeable Na+) | Sub-Surface Drainage (SSD) with corrugated perforated PVC pipes |
| **Physical Soil Structure** | Flocculated, good aeration, permeable | Dispersed, crusting, impermeable hard pan | Growing salt-tolerant crops: CSR-30 Basmati, KRL-210 Wheat |
| **Major Affected Districts** | Rohtak, Jhajjar, Hisar, Jind, Sonipat | Karnal, Kurukshetra, Kaithal, Panipat | Bio-drainage using Eucalyptus plantation along canal seepage |

ICAR-Central Soil Salinity Research Institute (CSSRI), Karnal (Est. 1969); Haryana Reclamation Corporation.

# River Systems of Haryana: Northern & Southern Drainage Networks

*Key Exam Takeaway: Haryana forms the great hydrological water divide between the Indus Basin and the Ganga Basin.*

## NORTHERN DRAINAGE SYSTEM (SHIVALIK ORIGIN)

• **Yamuna River:** The only perennial river; marks the entire eastern border with Uttar Pradesh; drains into the Bay of Bengal (Ganga System).

• **Ghaggar River:** Main seasonal river of northern Haryana; originates in Dagshai (HP); passes into Rajasthan desert (Indus / Inland System).

• **Markanda River:** Ancient Aruna; originates in Paonta/Dharti Dhar hills (HP); famous for seasonal fury and heavy siltation.

• **Tangri (Dangri) River:** Originates in Morni Hills; joins Markanda near Mullana (Ambala).

• **Saraswati River:** Ancient sacred river; originating near Adi Badri (Yamunanagar); revived by the Haryana Saraswati Heritage Development Board.

## SOUTHERN DRAINAGE SYSTEM (ARAVALLI ORIGIN)

• **Sahibi River:** Chief river of southern Haryana; originates in Kotputli/Bairath hills (Rajasthan); flows through Rewari and Gurugram into Najafgarh drain.

• **Krishnavati (Kasaunti) River:** Originates in Neem Ka Thana (Rajasthan); enters Mahendragarh; historically joins Sahibi.

• **Dohan River:** Originates near Mandoli (Rajasthan); mythological origin from the eye of Rishi Bhrigu's wife Divyapoloma; seasonal dry stream in Mahendragarh.

• **Indori River:** Originates in Nuh hills near old fort of Indore; divides into two branches near Rewari and meets Sahibi.

• **Inland Drainage:** Southern streams disappear into the sandy tracts of Mahendragarh and Rewari without reaching any sea.

# The Yamuna River: Entry, Course, Barrages & Basin Dynamics

*Key Exam Takeaway: Yamuna enters Haryana at Kalesar (Yamunanagar) and exits at Hasanpur (Palwal), spanning ~320 km.*

## RIVER COURSE & DEMARCATION

• **Glacial Source:** Originates from Yamunotri Glacier at Bandarpunch peak in Uttarkashi district (Uttarakhand) at ~6,387 m elevation.

• **Point of Entry:** Enters Haryana through Kalesar National Park forest in Yamunanagar district.

• **Natural Boundary Line:** Flows continuously southward defining the ~320 km interstate border between Haryana (Yamunanagar, Karnal, Panipat, Sonipat, Faridabad, Palwal) and Uttar Pradesh.

• **Point of Exit:** Leaves Haryana near Hasanpur village in Palwal district to flow into Mathura/Agra (UP).

• **Floodplain (Khadar):** Creates fertile flood-deposited newer alluvium belt across eastern Yamunanagar and Karnal.

## BARRAGES, DIVERSIONS & PROJECTS

• **Tajewala Barrage:** Historical stone diversion structure constructed by the British in 1873 to divert water into Western and Eastern Yamuna Canals.

• **Hathnikund Barrage:** Modern concrete barrage constructed in 1999 (operational 2002) 3 km upstream of Tajewala to handle super-floods (>8 lakh cusecs).

• **Okhla Barrage:** Located in Delhi/Faridabad border; feeds Agra Canal and Gurgaon Canal providing irrigation to southern districts.

• **Kishau, Renuka & Lakhwar Dams:** Upstream storage projects on tributaries (Tons, Giri, Yamuna) designed to ensure lean-season flow to Haryana.

• **FSO Ecological Note:** High industrial effluent contamination from Sonipat-Panipat-Gurugram drains (Dhanauri, Najafgarh) impacting downstream irrigation water.

# Ghaggar, Markanda (Aruna) & The Saraswati Heritage Project

*Key Exam Takeaway: Ghaggar creates the 'Naili' flood plain; Markanda merges into Saraswati at Sanisa Lake near Pehowa.*

| River Parameter | Ghaggar River | Markanda River (Aruna) | Saraswati River Project |
|---|---|---|---|
| **Origin Point** | Dagshai hills (Himachal Pradesh) at ~1,927m | Dharti Dhar / Paonta hills (HP) | Adi Badri foothills (Yamunanagar district) |
| **Entry into Haryana** | Near Kalka (Panchkula district) | Near Ambala district border | Adi Badri / Kathgarh cultural corridor |
| **Districts Traversed** | Panchkula, Ambala, Kaithal, Fatehabad, Sirsa | Ambala and Kurukshetra | Yamunanagar, Kurukshetra, Kaithal |
| **Major Tributaries** | Tangri, Markanda, Kaushalya (dam at Pinjore) | Runn, Begna, Nakti, and Tangri river | Chautang (ancient Drishadvati) historically |
| **Key Reservoirs / Sinks** | Ottu Barrage / Weir (Sirsa); dries in Thar Desert | Sanisa Lake (near Pehowa / Kurukshetra) | Palaeochannel excavations by HSHDB & ONGC |
| **Exam Terminology** | Flood plain called 'Naili'; non-perennial | Known as 'Aruna' in Mahabharata & Puranas | Mughalwali (Yamunanagar) site of active water revival |

Haryana Saraswati Heritage Development Board (HSHDB); State Remote Sensing Application Centre (HARSAC).

# Southern River Systems: Sahibi, Krishnavati, Dohan & Indori

*Key Exam Takeaway: Sahibi is the largest southern river, entering near Rewari and discharging into Najafgarh Jheel.*

## SAHIBI RIVER & MASANI BARRAGE

• **Chief Southern Artery:** Originates in Bairath / Sewar hills in Jaipur district, Rajasthan.

• **Entry into Haryana:** Enters Rewari district near Kotkasim border, flows through Pataudi (Gurugram) and Jhajjar.

• **Discharge Sink:** Empties into Najafgarh drain (historical Najafgarh lake) in Southwest Delhi.

• **Masani Barrage:** Built across Sahibi river in 1989 near Masani village (Rewari) on NH-48 for flood moderation and groundwater recharge.

• **Sub-Basin Significance:** Recharges vital deep aquifers in the water-stressed arid Rewari-Gurugram industrial belt.

## DOHAN, KRISHNAVATI & INDORI RIVERS

• **Dohan River:** Mythologically linked to Rishi Bhrigu's hermitage at Dhosi Hill; originates near Mandoli village; flows northward into Mahendragarh before getting lost in shifting sands.

• **Krishnavati (Kasaunti) River:** Originates from Dariba hills in Neem Ka Thana (Rajasthan); enters south of Narnaul; flows past Mahendragarh to disappear in Jhajjar depressions.

• **Indori River:** Takes birth near ancient ruined fort of Indore in Nuh (Mewat) hills; splits into two courses near Malab, one joining Sahibi near Rewari.

• **Ephemeral Nature:** All four streams are completely non-perennial and flow only during extreme cloudbursts in monsoon season.

• **Groundwater Depletion Alert:** Entire basin categorized as 'Over-exploited' / 'Dark Zone' by CGWB due to extensive tubewell extraction.

# Canal Irrigation Networks: Western Yamuna Canal & Bhakra System

*Key Exam Takeaway: WYC is Haryana's oldest canal (1355 CE); Tohana in Fatehabad is renowned as the 'City of Canals'.*

## WESTERN YAMUNA CANAL (WYC)

• **Historical Antiquity:** Originally excavated by Sultan Firoz Shah Tughlaq in 1355 CE to supply water to hunting lodges in Hisar.

• **Mughal Restoration:** Reconditioned by Akbar (1568) and remodeled as Nahr-i-Bihisht by Shah Jahan under Ali Mardan Khan to bring water to Delhi.

• **British Modernization:** Fully rebuilt in 1879 originating from Tajewala Barrage (now Hathnikund).

• **Branches & Command:** Main Branch divides into Delhi Branch, Hansi Branch, Bhalaut Branch, Barwala Branch, and Sunder Branch.

• **Command Districts:** Irrigates Yamunanagar, Karnal, Kaithal, Rohtak, Sonipat, Panipat, and Jind.

## BHAKRA SYSTEM & LIFT IRRIGATION SCHEMES

• **Bhakra Nangal Canal System:** Originates from Nangal Dam on Sutlej river; enters Haryana at Tohana (Fatehabad district) - known as 'Canal City'.

• **Main Branches:** Narwana Branch and Barwala Link Canal connecting Bhakra waters to Western Yamuna Canal system.

• **Jawaharlal Nehru (JLN) Lift Canal:** Multi-stage lift irrigation system engineered under CM Bansi Lal to pump water uphill to arid sandy terrains of Mahendragarh and Rewari.

• **Loharu (Indira Gandhi) & Siwani Lift Canals:** Lift canal systems designed to irrigate sand dune tracts of Bhiwani and Charkhi Dadri.

• **SYL Canal Dispute:** 214 km Sutlej-Yamuna Link canal (122 km in Punjab, 92 km in Haryana) constructed by Haryana but blocked by Punjab.

# Prominent Lakes, Wetland Reservoirs & Sacred Sarovars

*Key Exam Takeaway: Damdama is the largest natural lake; Brahma Sarovar is the sacred epicenter of Solar Eclipse pilgrimages.*

| Water Body / Sarovar | District Location | Origin / Structural Type | Key Exam Fact / Screening Point |
|---|---|---|---|
| **Damdama Lake** | Sohna (Gurugram) | Natural depression in Aravalli hills (Est. 1947) | Largest natural lake in Haryana (~3,000 acres); fed by rain runoff |
| **Badkhal Lake** | Faridabad | Man-made bundh constructed in 1947 | Built under CM Bansi Lal / Nehru for agricultural irrigation; name from 'Be-dakhal' |
| **Brahma Sarovar** | Thanesar (Kurukshetra) | Ancient sacred tank mentioned by Al-Biruni in 'Kitab-ul-Hind' | Largest man-made bathing tank in Asia (1800m x 1400m); Solar Eclipse dip |
| **Sannihit Sarovar** | Kurukshetra | Meeting point of 7 sacred Saraswatis | Believed to hold waters of all sacred teerthas on Amavasya; Lord Vishnu seat |
| **Karna Lake & Chakravarty** | Karnal (NH-44) | Artificial lake named after warrior Karna | Major roadside tourism hub of Haryana Tourism; Uchana complex |
| **Tilyar Lake** | Rohtak | Spread over 132 acres with Zoo | Major tourist complex and recreation hub of central Haryana |
| **Bhindawas & Khaparwas** | Jhajjar district | Freshwater wetland lake | Bhindawas is Haryana's largest wetland lake and designated Ramsar Site (2021) |
| **Blue Bird Lake** | Hisar | Artificial recreation lake & wetland | Famous migratory bird wetland near deer park and civil aerodrome |

# Climatic Zones, Temperature Extremes & Rainfall Isohyets

*Key Exam Takeaway: Climate is sub-tropical continental monsoon; Chhachhrauli is the 'Cherrapunji of Haryana' (>1100 mm).*

## TEMPERATURE & SEASONAL DYNAMICS

• **Classification:** Sub-tropical continental, semi-arid to sub-humid monsoon climate with extreme continental variations.

• **Temperature Extremes:** Searing summer peaks (>46°C - 48°C) in Hisar, Bhiwani, Sirsa, and Narnaul; freezing winter lows (<1°C - 2°C) with occasional ground frost (Pala/पाला) damaging potato and mustard crops.

• **Loo Winds (लू):** Hot, dry, dust-laden western winds blowing across southwestern plains from May to June.

• **Western Disturbances (Mawat / मावठ):** Winter precipitation in January-February brought by Mediterranean depressions; critical lifeline for Rabi wheat and gram.

• **HPSC Climate Question:** Southwestern fringe borders Thar Desert; northeastern fringe borders humid sub-Himalayas.

## RAINFALL DISTRIBUTION & ISOHYET MAP

• **Isohyet Gradient:** Average annual rainfall declines sharply from ~1,100 mm in the northeast to <300 mm in the southwest.

• **Highest Rainfall Belt:** Panchkula, Ambala, and Yamunanagar. Chhachhrauli tehsil (Yamunanagar) records the highest rainfall in the state (>1100 mm), earning the title 'The Cherrapunji of Haryana'.

• **Lowest Rainfall Belt:** Sirsa, Bhiwani, and Mahendragarh (average <300 mm; frequent drought episodes).

• **Monsoon Window:** South-West Monsoon (Arabian Sea and Bay of Bengal branches) active from early July to September, delivering ~80% of total annual rain.

• **Food Safety & Storage Implication:** High humidity (>80%) during July-August in eastern granary districts (Karnal, Kurukshetra) accelerates aflatoxin (Aspergillus flavus) proliferation in stored grains.

# Forest Cover Assessment: India State of Forest Report (ISFR)

*Key Exam Takeaway: Haryana has ~3.63% forest cover (1,603 sq km); Panchkula leads with 43.87% while Palwal/Fatehabad trail.*

| Forest Assessment Metric | Area (sq km) | Percentage of State Area | High-Yield Screening Exam Details |
|---|---|---|---|
| **Very Dense Forest (VDF)** | 28.00 sq km | 0.06% | Confined to Kalesar National Park (Yamunanagar) and high Morni ridgelines |
| **Moderately Dense Forest (MDF)** | 451.00 sq km | 1.02% | Distributed in Panchkula, Yamunanagar, Ambala, and Aravalli scrub tracts |
| **Open Forest (OF)** | 1,124.00 sq km | 2.54% | Largest forest category; agro-forestry belts, canal bundhs, roadside strips |
| **Total Recorded Forest Cover** | 1,603.00 sq km | 3.63% | One of India's lowest forest percentages; National Forest Policy target is 33% |
| **Tree Cover (Outside Recorded Forest)** | 1,425.00 sq km | 3.22% | Extensive Poplar and Eucalyptus farm forestry in Yamunanagar/Karnal |
| **Total Green Cover (Forest + Tree)** | 3,028.00 sq km | 6.85% | Combined forest and agro-forestry canopy in Haryana |
| **District with Highest Forest Cover** | Panchkula: 390.70 sq km | 43.87% | Leading district in absolute area and proportional percentage |
| **District with Lowest Forest Cover** | Palwal (13.7 sq km) & Fatehabad | 0.71% - 0.73% | Intensively cultivated agricultural plains with minimal natural canopy |

# National Parks of Haryana: Sultanpur (Ramsar) & Kalesar (Sal)

*Key Exam Takeaway: Haryana has exactly 2 National Parks: Sultanpur (Gurugram) and Kalesar (Yamunanagar).*

## 1. SULTANPUR NATIONAL PARK (GURUGRAM)

• **Geographical Location:** Farrukhnagar block, Gurugram district; spread over ~1.43 sq km (352 acres).

• **Historical Pioneer:** Declared a sanctuary in 1971 through efforts of famous British ornithologist Peter Scott and Dr. Salim Ali; upgraded to National Park in 1991.

• **Ramsar Site Recognition:** Designated as a Ramsar Wetland of International Importance in August 2021.

• **Avian Paradise:** Critical wintering ground for over 250 migratory bird species flying the Central Asian Flyway (Siberian crane, Greater flamingo, Bar-headed goose, Northern pintail).

• **Key Flora:** Acacia nilotica (Kikar), Ziziphus, Ber, and aquatic emergent marsh vegetation.

## 2. KALESAR NATIONAL PARK (YAMUNANAGAR)

• **Geographical Location:** Eastern foothills of Shivaliks, Yamunanagar district; spread over ~46.82 sq km (11,570 acres).

• **Upgraded to National Park:** December 2003 (surrounded by contiguous Kalesar Wildlife Sanctuary).

• **Interstate Forest Corridor:** Contiguous with Rajaji National Park (Uttarakhand) and Simbalbara National Park (Himachal Pradesh).

• **Dominant Flora:** Pure stands of mature Sal (Shorea robusta) forest, Semul, Khair, and Shisham; the only natural Sal forest in Haryana.

• **Faunal Diversity:** Wild Asian elephants (visiting from Rajaji), Leopards, Barking deer, Sambar, Goral, King Cobra, and Red Junglefowl.

• **Historical Monument:** Ancient British-era Kalesar Dak Bungalow (1901) and historical Shiva temple.

# Wildlife Sanctuaries of Haryana: 8 Sanctuaries & 2 Reserves

*Key Exam Takeaway: Abubshahar (Sirsa) is largest (11,530 ha); Chhilchhila (Kurukshetra) is smallest (28 ha); Bhindawas is Ramsar.*

| Sanctuary / Reserve | District Location | Area (Hectares) | Key Protected Fauna & Exam Points |
|---|---|---|---|
| **1. Abubshahar Sanctuary** | Sirsa (Mandi Dabwali border) | 11,530 ha | Largest sanctuary in Haryana; prime habitat for Blackbuck (Kala Hiran) & Nilgai |
| **2. Bhindawas Sanctuary** | Jhajjar district | 411.55 ha | Man-made freshwater wetland; Ramsar Site (2021); ~250 bird species; largest wetland lake |
| **3. Khaparwas Sanctuary** | Jhajjar (1.5 km from Bhindawas) | 82.70 ha | Wetland sanctuary recognized for waterfowl protection; notified 1991 |
| **4. Kalesar Wildlife Sanctuary** | Yamunanagar district | 5,435 ha | Adjoins Kalesar NP; dense Sal & mixed forest; Leopard, Wild Boar, Chital corridor |
| **5. Bir Shikargah Sanctuary** | Pinjore (Panchkula district) | 767.30 ha | Houses the famous Vulture Conservation Breeding Centre (VCBC); Sambar, Barking Deer |
| **6. Khol Hi-Raitan Sanctuary** | Panchkula (Shivalik belt) | 4,883 ha | Critical Shivalik eco-fragile corridor connecting Bir Shikargah and Morni Hills |
| **7. Nahar Wildlife Sanctuary** | Kosli tehsil (Rewari district) | 211.35 ha | Famous for Blackbuck, Jackal, Fox, and Peafowl in dry scrub deciduous forest |
| **8. Chhilchhila Sanctuary** | Kurukshetra (near Pehowa) | 28.92 ha | Smallest wildlife sanctuary in Haryana; vital shallow lake nesting site for winter ducks |
| **Conservation Reserves** | Bir Bara Ban (Jind) & Saraswati (Kaithal/Kurukshetra) | Bir Bara Ban: 419 ha<br>Saraswati: 4,452 ha | Bir Bara Ban (Jind town green lung); Saraswati Reserve (Bir Sonty) 2nd largest forest patch |

# Specialized Wildlife Breeding Centres: Location & Species Focus

*Key Exam Takeaway: Pinjore hosts Asia's first Jatayu Vulture Conservation Breeding Centre (VCBC).*

## AVIAN BREEDING HUBS (PANCHKULA)

• **Jatayu Vulture Conservation Breeding Centre (VCBC):** Pinjore, Panchkula (Est. 2001 in collaboration with BNHS). Pioneer in captive breeding of 3 critically endangered Gyps species: Oriental white-backed, Long-billed, and Slender-billed vultures.

• **Diclofenac Ban Catalyst:** Research at Pinjore proved veterinary non-steroidal anti-inflammatory drug (NSAID) Diclofenac caused catastrophic visceral gout and renal failure in vultures, leading to a nationwide ban.

• **Pheasant Breeding Centre:** Morni, Panchkula district (Est. 1992-93). Specializes in Red Junglefowl and Kalij pheasant conservation.

• **Red Junglefowl Genetic Purity:** Rigorous DNA testing to prevent hybridization with domestic poultry.

## FAUNAL BREEDING & REPTILE CENTRES

• **Chinkara Breeding Centre:** Kairu, Bhiwani district. Established for breeding Indian Gazelle (Gazella bennettii) adapted to dry sand dune habitats.

• **Peacock & Chinkara Breeding Centre:** Jhabua, Rewari district. Dedicated to Indian Peafowl (Pavo cristatus) and Chinkara protection in Aravalli scrub.

• **Crocodile Breeding Centre (Bhor Saidan):** Bhor Saidan village near Pehowa, Kurukshetra district (Est. 1981-82). Houses Mugger crocodile (Crocodylus palustris).

• **Blackbuck Breeding Centre:** Pipli Mini Zoo, Kurukshetra and Rohtak Zoo.

• **Mini Zoos of Haryana:** Pipli Zoo (Kurukshetra), Rohtak Zoo, and Bhiwani Zoo under CZA regulatory standards.

# Aravalli Green Wall Project, Urban Oxy-Vans & Climate Action

*Key Exam Takeaway: Aravalli Green Wall Project aims to afforest a 5 km buffer along the 1,400 km Aravalli belt.*

## ARAVALLI GREEN WALL & ECO-RESTORATION

• **Aravalli Green Wall Initiative:** Inaugurated at Tikli village in Gurugram on International Day of Forests (March 2023).

• **National Buffer Scale:** 1,400 km long and 5 km wide green belt covering Haryana, Rajasthan, Gujarat, and Delhi.

• **Core Objectives:** Arrest eastward sand ingress from Thar desert; rejuvenate degraded Aravalli water recharge zones; combat regional desertification under UNCCD commitments.

• **Species Selection:** Indigenous drought-resilient trees: Kikar, Dhak (Butea monosperma), Ronjh, Neem, Peepal, and wild Ber.

• **Aravalli Safari Park:** Planned 10,000-acre project in Gurugram and Nuh districts; slated as the world's largest curated safari outside Africa.

## OXY-VANS, PRAN VAYU PENSION & STUBBLE MANAGEMENT

• **Oxy-Van (ऑक्सी वन) Initiative:** Urban micro-forests conceptualized during post-COVID environmental push. Flagship 800-acre Oxy-Van established at Karnal along Western Yamuna Canal.

• **Panchkula Oxy-Van:** Bir Ghaggar forest patch developed with 7 dedicated thematic groves (Panchavati, Navagraha, Nakshatra, Sugandh Vatika).

• **Pran Vayu Devta Pension Scheme:** Unique state scheme providing annual lifelong pension (~Rs 2,750+) to caretakers of healthy trees older than 75 years.

• **Air Quality Action (CAQM):** Stringent Commission for Air Quality Management regulations in 14 NCR districts of Haryana regulating stubble burning (Parali / पराली) and industrial boilers.

• **Bio-Energy Policy:** Extensive setup of 2G Ethanol Plant by IOCL at Panipat utilizing paddy straw to tackle seasonal smog.

# Vedic Civilization: Brahmavarta, Saraswati Basin & Kurukshetra

*Key Exam Takeaway: Manu Smriti defines Brahmavarta as the land between sacred rivers Saraswati & Drishadvati.*

## BRAHMAVARTA & VEDIC GEOGRAPHY

• **Brahmavarta (ब्रह्मावर्त):** Defined in Manu Smriti as the holy tract created by the gods between the divine rivers Saraswati and Drishadvati (Chautang).

• **Rigvedic Composition:** The majority of Rigvedic hymns were composed by Vedic seers (Bharadwaja, Vashistha, Vishvamitra) on the banks of the sacred Saraswati river.

• **Brahmarshi Desha:** Tract including Kurukshetra, Matsya, Panchala, and Surasena regarded as the highest ethical standard.

• **Kurukshetra Naming:** Named after King Kuru of Bharata dynasty, who ploughed the field with a golden plough across 48 kos (48 Kos Parikrama) to cultivate eight moral virtues (Satya, Dharma, Shaucha, etc.).

• **Jyotisar (ज्योतिसर):** The sacred banyan tree site where Lord Krishna delivered the celestial message of the Bhagavad Gita to Arjuna before the 18-day Mahabharata War.

## EPIC MAHABHARATA LANDSCAPE

• **48 Kos Parikrama:** Circumambulation encompassing over 134 sacred tirthas spread across Kurukshetra, Kaithal, Karnal, and Jind districts.

• **Karnal Connection:** Named after Anga-raj Karna, famous for unshakeable charity; Karna Lake and Karna Kot.

• **Prithudaka (Pehowa):** Ancient pilgrimage town on Saraswati river named after King Prithu; premier center in northern India for Pitru Tarpan / ancestral shradh rites.

• **Ban Ganga (Narkatari):** Site near Kurukshetra where Arjuna shot an arrow into the earth to quench Bhishma Pitamah's thirst on the bed of arrows.

• **Abhimanyu Pur (Amin):** Ancient village where 16-year-old warrior Abhimanyu was trapped in the Chakravyuha; recently renamed Abhimanyupur.

Rigveda; Manu Smriti (Chapter 2, Verse 17); Archaeological Survey of India (ASI).

# Rakhigarhi (Hisar): Largest Harappan Metropolis in South Asia

*Key Exam Takeaway: Spanning ~550 hectares across 11 mounds, Rakhigarhi is larger than Mohenjo-daro.*

## EXCAVATION HISTORY & URBAN SCALE

• **Geographical Setting:** Located in Narnaund tehsil of Hisar district in the dry palaeochannel of the ancient Drishadvati (Chautang) river.

• **Discovery & Excavations:** First identified by Acharya Bhagwan Dev (1960s); extensively excavated by ASI under Dr. Amarendra Nath (1997-2000) and Prof. Vasant Shinde (Deccan College, 2011-2017).

• **Magnitude & Scale:** Comprises 11 distinct archaeological mounds covering ~550 hectares, officially surpassing Mohenjo-daro and Harappa as the largest Mature Harappan site.

• **Urban Masterplanning:** Mud-brick defensive perimeter walls, multi-roomed houses with planned bathrooms, baked brick soak pits, covered street drainage, and terracotta drainage pipes.

• **Mound 7 Cemetery:** Yielded over 60 burials (primary, secondary, and symbolic burials) with rich grave pottery and jasper/steatite beads.

## SCIENTIFIC DISCOVERIES & DNA BREAKTHROUGH

• **Landmark aDNA Discovery (Cell, 2019):** Ancient DNA extracted from female skeleton (I6113) at Mound 7 proved: Indigenous South Asian ancestry with zero Steppe pastoralist or Anatolian farmer genetic contribution.

• **Monumental Granary:** Massive multi-chambered mud-brick structure with lime and decomposed grass plaster, engineered for grain storage and preservation against moisture.

• **Jewellery Workshop:** Uncovered lapidary beads of carnelian, agate, jasper, gold foils, copper mirrors, and micro-steatite beads indicating major trade hub.

• **Gold Smelting & Metallurgy:** Furnaces with crucibles containing minute traces of refined gold and copper slag.

• **World-Class Site Museum:** Central ASI and State Government site museum with on-site hostel, interpretation center, and experiential gallery.

Archaeological Survey of India (Dr. Amarendra Nath & Prof. Vasant Shinde); Cell / Science DNA Studies (2019).

# Banawali, Bhirrana & Kunal: The Fatehabad Civilizational Cradle

*Key Exam Takeaway: Bhirrana is India's oldest Harappan site (~7500 BCE); Banawali yielded clay plough & barley.*

| Archaeological Site | District & River Basin | Chronological Antiquity | Key Excavated Artefacts & Screening Traps |
|---|---|---|---|
| **Bhirrana (भीड़ाना)** | Fatehabad (Ancient Saraswati bed) | Hakra Ware phase (c. 7500 BCE - 6000 BCE) | Oldest Harappan settlement in South Asia (C-14 dating); red ware pot shard featuring graffito of 'Dancing Girl' mimicking Mohenjo-daro bronze idol |
| **Banawali (बनावली)** | Fatehabad (14 km from Fatehabad town) | Pre-Harappan, Mature, and Post-Harappan | Excavated by Dr. R.S. Bisht (1974); terracotta toy plough (मिट्टी का हल); rich caches of barley grains (जौ); radial town planning layout without rigid grid |
| **Kunal (कुणाल)** | Fatehabad (Ratia tehsil on Saraswati) | Early/Pre-Harappan (c. 4000 BCE) | Excavated by J.S. Khatri & M. Acharya; two exquisite royal silver tiaras/crowns (चांदी के मुकुट); gold disc ornaments; bone tools; pit-dwellings preceding brick houses |
| **Agroha (अग्रोहा)** | Hisar (NH-9 highway) | Post-Harappan to Kushan/Gupta period | Capital of legendary King Agrasen (Agrawal community); coins bearing legend 'Agrodaka'; terracotta tablet with 7 musical notes (Sa Re Ga Ma Pa Dha Ni) |

# Mitathal, Siswal, Farmana (Daksh Khera) & Balu

*Key Exam Takeaway: Farmana revealed 70 Harappan burials; Balu yielded earliest evidence of garlic cultivation.*

## BHIWANI & HISAR PROTO-HISTORIC NETWORK

• **Mitathal (Bhiwani):** Excavated by Dr. Suraj Bhan (1968). Yielded Late Harappan copper hoard including copper axes, parashu, and rings; shows transition to Ochre Coloured Pottery (OCP).

• **Siswal (Hisar):** Excavated by Dr. Suraj Bhan (1970). Type-site for the distinct 'Siswal Culture' characterized by Kalibangan-I related painted pottery in non-alluvial dry zones.

• **Tigrana (Bhiwani):** Recent excavations revealing large-scale terracotta and semi-precious stone bead manufacturing workshop.

• **Naurangabad (Bhiwani):** Ancient Prakritnaket; coins, seals, and terracotta moulds of the Yaudheya Republic (योध्येय गणराज्य) bearing legend 'Yaudheya Ganasya Jaya'.

## FARMANA, BALU & AGRICULTURAL ANTIQUITY

• **Farmana / Daksh Khera (Rohtak):** Joint excavation by Deccan College (Prof. Vasant Shinde) and Research Institute for Humanity and Nature (Japan). Massive 70-grave cemetery, copper ornaments, and jasper stamp seals.

• **Balu (Kaithal):** Excavated by Kurukshetra University (Dr. U.V. Singh). Crucial archaeobotanical find: Earliest evidence of cultivated Garlic (Allium sativum), green gram, and mustard seeds.

• **Sugh / Shrughna (Yamunanagar):** Ancient fortified port city on Yamuna; punch-marked coins, terracotta figurines, and Ashokan Buddhist heritage.

• **Bhadra & Daulatpur (Kurukshetra):** Late Harappan sites yielding painted grey ware (PGW) and clay seals.

# The Topra Ashokan Pillar, Chaneti Stupa & Buddhist Heritage

*Key Exam Takeaway: Firoz Shah Tughlaq moved the Topra Ashokan Pillar (Yamunanagar) to Kotla, Delhi in 1356 CE.*

## TOPRA ASHOKAN PILLAR EDICTS

• **Topra Kalan (Yamunanagar):** Original site of Emperor Ashoka's monumental monolithic polished sandstone pillar (3rd century BCE) bearing 7 Major Pillar Edicts in Brahmi script.

• **Only Pillar with 7 Edicts:** Topra is the only Ashokan pillar containing the entire Seventh Pillar Edict summarizing Dhamma policy (other pillars contain only 6).

• **Transport to Delhi (1356 CE):** Sultan Firoz Shah Tughlaq transported this multi-ton pillar on a 42-wheeled carriage pulled by 8,400 men and floated it down the Yamuna on large boats to install it atop Firoz Shah Kotla, Delhi.

• **Topra Ashoka Edicts Park:** Newly established memorial park in Yamunanagar housing exact replicas of all 14 Ashokan Rock Edicts and Topra pillar replica.

## CHANETI STUPA & SHRUGHNA MONASTERIES

• **Chaneti Buddhist Stupa (Yamunanagar):** Located 3 km from Jagadhri; built in 3rd century BCE during Ashoka's reign using concentric baked bricks, mimicking Sanchi hemisphere style.

• **Sugh (Ancient Shrughna):** Major Buddhist monastery mentioned by Chinese pilgrim Xuanzang (Hiuen Tsang); recorded 5 monasteries with over 700 Buddhist monks in Shrughna.

• **Karnal Inscription:** Kharosthi script inscription discovered at Karnal dating to 1st century BCE / CE.

• **Adi Badri Stupa:** Excavated Buddhist monastery and stupa structures in Yamunanagar foothills highlighting cultural confluence.

• **HPSC History Trap:** Remember, Topra was originally in Ambala district; after 1989 reorganization, it falls squarely in Yamunanagar district.

# Pushyabhuti Dynasty: Harshavardhana, Thanesar & Xuanzang

*Key Exam Takeaway: Harsha ruled from Thanesar (Kurukshetra) before shifting his imperial capital to Kannauj.*

## PUSHYABHUTIS OF THANESAR

• **Dynasty Founder:** Pushyabhuti (worshipper of Lord Shiva) founded dynasty at Sthanishvara (modern Thanesar in Kurukshetra).

• **Prabhakarvardhana:** First powerful king; adopted title 'Paramabhattaraka Maharajadhiraja'; called 'Huna-harina-kesari' (a lion to the Huna deer).

• **Accession of Harsha (606 CE):** Ascended throne at age 16 following the murder of elder brother Rajyavardhana by Gauda King Shashanka.

• **Shifting Capital:** Unified crowns of Thanesar and Maukhari (Kannauj) and shifted imperial capital to Kannauj for administrative centrality.

• **Literary Sovereign:** Harsha himself authored three acclaimed Sanskrit plays: Ratnavali, Priyadarsika, and Nagananda.

## COURT CULTURE & ARCHAEOLOGICAL LEGACY

• **Banabhatta (Court Poet):** Authored 'Harshacharita' (first historical biography in Sanskrit) and 'Kadambari'; described the prosperity of Srikantha Janapada (Haryana).

• **Visit of Xuanzang (Hiuen Tsang):** Traveled across Haryana (634-644 CE); stayed in Thanesar; praised social harmony, agricultural fertility, and Buddhist viharas.

• **Defeat by Pulakeshin II:** Harsha's southward expansion was halted at the Narmada River by Chalukya king Pulakeshin II (recorded in Aihole Inscription by Ravikirti).

• **Harsha-ka-Tila (Thanesar):** Excavated multi-period cultural mound in Thanesar revealing Kushana, Gupta, Harsha, Rajput, and Sultanate architectural layers.

• **Sheikh Chilli's Tomb:** Adjacent Sufi monument in Thanesar often called 'Taj Mahal of Haryana'.

# Tomar Dynasty, Prithviraj Chauhan & The Battles of Tarain (1191-92)

*Key Exam Takeaway: First Battle of Tarain (1191): Prithviraj won; Second Battle (1192): Ghori won, founding Delhi Sultanate.*

| Historical Event / Dynasty | Year / Period | Protagonists / Builders | Key Screening Points & Monuments |
|---|---|---|---|
| **Tomar Dynasty Rule** | 8th - 12th Century CE | Anangpal Tomar I & II | Founded 'Dhillika' (Delhi); built Surajkund Sun Temple & Dam in Faridabad; Pehowa Inscription of Bhoja |
| **Asigarh Fort (Hansi)** | 12th Century CE | Prithviraj Chauhan III | Also called 'Sword Fort' (Asi = Sword); strategic military stronghold of Chauhan kingdom |
| **First Battle of Tarain** | 1191 CE (Tarori, Karnal) | Prithviraj Chauhan vs Muhammad Ghori | Prithviraj's Rajput confederacy routed Ghori; Govind Rai wounded Ghori; Ghori fled to Ghazni |
| **Second Battle of Tarain** | 1192 CE (Tarori, Karnal) | Muhammad Ghori vs Prithviraj Chauhan | Ghori used 10,000 mounted archers; Rajput forces defeated; Prithviraj captured near Saraswati river; paved way for Muslim rule |
| **Qutb-ud-din Aibak Campaign** | 1192 - 1206 CE | First Mamluk Sultan | Crushed local resistance in Hansi, Kohram, and Rewari; established Turkish military outposts |

Minhaj-i-Siraj's 'Tabaqat-i-Nasiri'; Chand Bardai's 'Prithviraj Raso'; ASI Monuments.

# Delhi Sultanate: Razia Sultan at Kaithal, Balban & Firoz Shah

*Key Exam Takeaway: Razia Sultan was assassinated near Kaithal (1240 CE); Firoz Shah Tughlaq founded Hisar & Fatehabad.*

## RAZIA SULTAN, BALBAN & TIMUR'S SCOURGE

• **Razia Sultan's Assassination (14 Oct 1240):** Razia (first female Muslim monarch of Delhi) and husband Malik Altunia were captured and murdered by rebel forces near Kaithal; historical Tomb of Razia Sultan exists in Kaithal town.

• **Ghiyas-ud-din Balban:** Subdued ferocious Mewati guerrilla rebels in southern Haryana; deforested Gurgaon/Mewat forests around Delhi; built a cordon of frontier forts at Hansi, Rewari, and Narnaul.

• **Invasion of Timur (1398 CE):** Plundered Sirsa, Fatehabad, Aharwan, Tohana, and Asandh en route to Delhi; faced fierce resistance from Jat and Ahir peasants.

• **Yaudheya & Khanzada Chiefs:** Semi-autonomous regional chieftains maintaining de facto control over south Haryana countryside.

## FIROZ SHAH TUGHLAQ'S URBAN & CANAL ARCHITECTURE

• **Foundation of Hisar-e-Firoza (1354 CE):** Built by Sultan Firoz Shah Tughlaq as a fortified royal township; features Gujri Mahal (built for his beloved milkmaid Gujri), Lat ki Masjid, and Diwan-e-Aam.

• **Lat (Pillar) of Firoz Shah:** Ashokan-style sandstone monolithic pillar section erected in the courtyard of Hisar Fort complex.

• **Foundation of Fatehabad:** Founded in 1351 CE and named after his newborn son Fateh Khan; houses Humayun Mosque and an Ashokan pillar fragment.

• **Firozpur Jhirka:** Established in southern Mewat hills as an administrative post.

• **Agricultural Canal Legacy:** Excavated original Western Yamuna Canal to supply water from Yamuna river to royal gardens in Hisar.

# The Three Battles of Panipat (1526, 1556, 1761): Strategic Analysis

*Key Exam Takeaway: Panipat altered Indian history thrice: 1526 (Mughal advent), 1556 (Mughal restoration), 1761 (Maratha decimation).*

| Parameter | First Battle of Panipat | Second Battle of Panipat | Third Battle of Panipat |
|---|---|---|---|
| **Date of Conflict** | 21 April 1526 | 5 November 1556 | 14 January 1761 (Makar Sankranti) |
| **Opposing Forces** | Zahir-ud-din Babur vs Sultan Ibrahim Lodi | Akbar (Bairam Khan) vs Hemu (Vikramaditya) | Ahmad Shah Abdali vs Sadashivrao Bhau (Marathas) |
| **Military Innovation** | Tulughma flanking tactic & Rumi cart-gun carriage (Araba) | Artillery bombardment; stray arrow hit Hemu's eye while atop 'Hawai' elephant | Abdali's mobile swivel camel-mounted guns (Zamburak) & cavalry |
| **Casualties / Decisive Event** | Ibrahim Lodi killed on battlefield (Tomb in Panipat) | Hemu captured unconscious & beheaded by Bairam Khan | 28,000+ Marathas slain; Vishwasrao & Jankoji Scindia martyred |
| **Historical Outcome** | Fall of Delhi Sultanate; foundation of Mughal Empire | Decisive consolidation of Mughal rule under Akbar | Shattered Maratha dream of Pan-Indian rule; paved way for British |
| **Memorial Landmark** | Kabuli Bagh Mosque (built by Babur in Panipat) | Hemu's Samadhi Sthal at Shodapur village | Kala Amb (Black Mango Tree) Memorial in Panipat |

Baburnama; Abul Fazl's 'Akbarnama'; J.N. Sarkar's 'Fall of the Mughal Empire'; Panipat Museum.

# Hemu (Samrat Hemchandra Vikramaditya): Rise from Rewari to Delhi

*Key Exam Takeaway: Hemu won 22 consecutive battles and was crowned Emperor at Purana Qila, Delhi (Oct 7, 1556).*

## ASCENT OF A NATIVE HERO

• **Humble Beginnings:** Born in a Brahmin/Dhussar merchant family at Machhari village (Alwar) and brought up in Qutabpur, Rewari.

• **Mercantile to Military Genius:** Started as a saltpetre trader (vital ingredient for gunpowder); recognized for brilliant logistics by Sultan Islam Shah Suri.

• **Prime Minister & Commander-in-Chief:** Appointed Wazir and supreme military general by Sultan Adil Shah Suri; won 22 consecutive battles against Afghan rebels and Mughal commanders.

• **Capture of Delhi & Agra (1556):** Smashed Mughal Governor Tardi Beg Khan at the Battle of Tughlaqabad (Delhi) on 6 October 1556.

• **Imperial Coronation:** Assumed imperial title 'Samrat Hemchandra Vikramaditya' at Purana Qila, Delhi on 7 October 1556; re-established indigenous Hindu sovereignty.

## BATTLE OF PANIPAT II & HISTORICAL LEGACY

• **The Tragic Turn at Panipat:** On 5 November 1556, Hemu's forces were routing the Mughal army under Bairam Khan when a rogue stray arrow pierced Hemu's right eye, causing him to fall unconscious in his howdah.

• **Mughal Vengeance:** Captured unconscious and brought before the 13-year-old Akbar; beheaded by Bairam Khan; Hemu's head was sent to Kabul and torso hung outside Delhi gate.

• **Father's Martyrdom:** Hemu's 80-year-old father Puran Das refused to convert and was martyred at Alwar.

• **Historical Stature:** Described by historians as the 'Napoleon of Medieval India' and the last Hindu sovereign to hold Delhi's imperial throne.

• **Memorial:** Hemu's Samadhi Sthal situated at Shodapur village near Panipat.

Vincent Smith, 'Akbar the Great Mogul'; K.K. Bhardwaj, 'Hemu: Napoleon of Medieval India'.

# Battle of Karnal (1739) & The Adventurer Raja: George Thomas at Hansi

*Key Exam Takeaway: Nadir Shah defeated Muhammad Shah at Karnal; Irish adventurer George Thomas ruled Hansi (1797-1801).*

## BATTLE OF KARNAL (1739) & COLLAPSE

• **Battle Date:** 24 February 1739 at Karnal plains.

• **Opponents:** Nadir Shah (Shah of Persia / Afsharid dynasty) vs Mughal Emperor Muhammad Shah 'Rangeela'.

• **Military Disaster:** In just 3 hours of combat, Nadir Shah's disciplined musketeers annihilated the superior-numbered Mughal army; Burhan-ul-Mulk captured.

• **Plunder of Delhi:** Nadir Shah marched to Delhi, ordered horrific general massacre (Qatl-e-Aam), and looted the Peacock Throne (Takht-e-Taus) and the Koh-i-Noor diamond.

• **Treaty of Surji-Anjangaon (30 Dec 1803):** Following Second Anglo-Maratha War, Daulat Rao Scindia ceded Haryana territory to British East India Company.

## GEORGE THOMAS: THE JAHAZI SAHIB OF HANSI

• **George Thomas ('Jahazi Sahib'):** Irish deserter who rose through mercenary ranks under Begum Samru of Sardhana and Maratha commander Appa Khande Rao.

• **Independent Kingdom at Hansi (1797-1801):** Established independent rule carving out 800 villages; rebuilt Asigarh Fort at Hansi as his royal capital.

• **Mint & Artillery Foundry:** Established his own currency mint issuing 'Sikka-e-Sahib' coins and cast European-style cannons in Hansi.

• **Downfall:** Besieged by French General Pierre Perron commanding Maratha forces; surrendered Hansi in 1802; died en route to Calcutta at Berhampore.

• **Hansi Heritage:** George Thomas's residence 'Jahaz Kothi' (so called because it resembled a ship) preserved as heritage monument in Hisar.

William Irvine's 'Later Mughals'; Capt. Francklin's 'Military Memoirs of Mr. George Thomas'.

# The 1857 Uprising: Outbreak at Ambala Cantonment (10 May 1857)

*Key Exam Takeaway: The uprising broke out at Ambala Cantt 9 hours BEFORE Meerut (8-9 hours earlier on 10 May 1857).*

## AMBALA CANTT: THE FORGOTTEN EPICENTER

• **Pioneering Revolt Timing:** Mutiny broke out in the 5th and 60th Native Infantry at Ambala Cantonment at ~9:00 AM on Sunday, 10 May 1857 (almost 9 hours prior to the Meerut rebellion at 5:00 PM).

• **The Traitor Sham Singh:** Sepoy Sham Singh betrayed the mutineers by leaking plan details to British officers, allowing commander General Anson to disarm the regiments.

• **Ambala War Memorial:** Grand 22-acre international-standard memorial museum constructed on NH-44 at Ambala Cantt to honor unsung 1857 martyrs.

• **Strategic Ambala Cantt:** Established in 1843 (transferred from Karnal after malaria epidemic); northern military headquarters of British India.

## DISTRICT-WIDE SPREAD OF RESISTANCE

• **Regional Flare-Ups across Haryana:**

• • **Rohtak & Jhajjar:** Rebel sepoys led by Risaldar Bisarat Ali routed British district officers.

• • **Hisar & Hansi:** Led by Muhammad Azim (Prince of Delhi) and Hukam Chand Jain; British deputy commissioner John Wedderburn killed.

• • **Sirsa:** 4th Irregular Cavalry rebelled; British fled to Bikaner.

• • **Panipat:** Led by Imam Ali Qalandar and local Gujjar/Jat villagers cutting Grand Trunk Road communication.

• • **Mewat / Gurgaon:** Sadruddin Mewati led thousands of Meos against British forces; burnt police outposts.

# Key Leaders of 1857: Rao Tula Ram, Raja Nahar Singh & Abdul Rahman

*Key Exam Takeaway: Rao Tula Ram liberated Rewari; Raja Nahar Singh ('Lion of Ballabhgarh') was hanged in Chandni Chowk.*

| 1857 Rebel Leader | Region / Princely State | Resistance Milestones | Fate & Commemoration |
|---|---|---|---|
| **Rao Tula Ram (with cousin Gopal Dev)** | Rewari & Ahirwal region | Liberated Rewari; established cannon factory at Rampura Fort; funded Mughal treasury at Delhi | Escaped to Iran/Afghanistan to seek Russian aid; died in Kabul (23 Sep 1863); Haryana Shaheedi Diwas |
| **Raja Nahar Singh ('Ballabhgarh Sher')** | Ballabhgarh (Faridabad) | Guarded Delhi's southern frontier; defied British commands; protected refugees | Arrested deceitfully; publicly hanged at Chandni Chowk on 9 Jan 1858; Ballabhgarh metro named in honor |
| **Nawab Abdul Rahman Khan** | Jhajjar princely state | Mobilized cavalry and infantry in support of Emperor Bahadur Shah Zafar | Tried by military commission; hanged outside Red Fort on 23 Dec 1857; Jhajjar territory annexed |
| **Nawab Ahmad Ali** | Farrukhnagar (Gurugram) | Co-ordinated Meo and Ahir peasant resistance against British relief columns | Hanged at Kotwali Chandni Chowk on 23 Jan 1858 |
| **Lala Hukam Chand Jain & Muneer Beg** | Hansi (Hisar district) | Drafted plans to overthrow British garrisons across western Haryana | Hanged simultaneously from the same tree in Hansi in 1858 |

National Archives of India; Punjab Mutiny Records (1857-58); Haryana Sahitya Akademi.

# Battle of Naseepur (16 Nov 1857) & Haryana Shaheedi Diwas (23 Sep)

*Key Exam Takeaway: Haryana observes Shaheedi Diwas on 23 September in sacred memory of Rao Tula Ram.*

## THE BLOODBATH AT NASEEPUR

• **Battlefield Setting:** Naseepur plain situated 3 km north of Narnaul town (Mahendragarh district).

• **Date of Epic Battle:** 16 November 1857.

• **Allied Forces:** Joint patriotic command under Rao Tula Ram, Gopal Dev, Gen. Abdus Samad Khan (Jhajjar), and Jodhpur Legion mutineers (5,000+ fighters).

• **Opposing British Column:** Col. Gerrard leading combined force of 1st European Bengal Fusiliers, Sikh cavalry (Patiala/Jind), and Multani horse.

• **Battle Climax:** Fierce hand-to-hand combat; British Commander Col. Gerrard was shot dead on the field; Captain Morehead wounded; allied forces eventually succumbed to heavy artillery fire.

## SHAHEEDI DIWAS & TERRITORIAL REPERCUSSIONS

• **Haryana Shaheedi Diwas (23 September):** Observed as a gazetted State public holiday every year in Haryana to honor Rao Tula Ram who breathed his last in Kabul on 23 September 1863.

• **Post-1857 Disenfranchisement:** Under the Queen's Proclamation of 1858, Haryana was ripped away from North-Western Provinces and merged with Punjab as collective punishment.

• **Princely Rewards for Loyalist Chiefs:** Princely states that aided the British (Patiala, Nabha, Jind, Kapurthala) were rewarded with confiscated territories of Jhajjar, Dadri, and Narnaul.

• **Exemplary Ahirwal Valour:** Legacy continued through the Battle of Rezang La (1962) where 13 Kumaon (Ahir troops) made supreme sacrifice under Major Shaitan Singh.

# Sir Chhotu Ram (Deenbandhu / Rahbar-i-Azam): Peasant Legislation

*Key Exam Takeaway: Sir Chhotu Ram passed revolutionary agrarian debt cancellation acts & conceptualized Bhakra Dam.*

## BIOGRAPHICAL & POLITICAL LANDMARKS

• **Early Life & Identity:** Born Ram Richhpal on 24 November 1881 at Garhi Sampla (Rohtak district); earned law degree from Agra College.

• **Jat Gazette (1916):** Launched premier weekly newspaper 'Jat Gazette' from Rohtak in Urdu/Hindi to articulate rural peasant grievances.

• **Unionist Party (Zamindara League 1923):** Founded non-communal agrarian party with Sir Fazl-i-Hussain representing Hindu, Sikh, and Muslim agriculturists.

• **Bhakra Dam Visionary:** Spearheaded water agreement with the Raja of Bilaspur (1944) laying the legal and financial foundation for Bhakra Dam.

• **Iconic Statues:** Prime Minister Narendra Modi unveiled a monumental 64-foot statue of Sir Chhotu Ram at Garhi Sampla in 2018.

## LEGISLATIVE MAGNA CARTA FOR FARMERS

• **Landmark Pro-Farmer Legislative Acts:**

• **1. Punjab Relief of Indebtedness Act 1934:** Liquidated predatory moneylender debts; capped compound interest; protected oxen and agricultural tools from auction.

• **2. Punjab Debtor's Protection Act 1936:** Exempted ancestral agricultural land and standing crops from court attachment.

• **3. Punjab Restitution of Mortgaged Lands Act 1938:** Restored hundreds of thousands of mortgaged agricultural acres back to destitute farmers without cash payment.

• **4. Punjab Agricultural Produce Markets Act 1939:** Established institutional Mandis (Market Committees) eliminating fraudulent merchant weights (Dami).

• **Titles:** Conferred 'Rao Bahadur' (1919), Knighthood ('Sir' in 1937), 'Rahbar-i-Azam' by grateful peasants.

Dr. Raghuvendra Tanwar, 'Sir Chhotu Ram: A Biography'; Punjab Legislative Assembly Debates.

# Freedom Struggle Stalwarts: Lala Lajpat Rai & Pandit Neki Ram Sharma

*Key Exam Takeaway: Pt. Neki Ram Sharma founded 'Sandesh'; declared 'I want the entire land of India'.*

| Freedom Fighter / Pioneer | Native Region | Major Publications / Institutions | Iconic Contribution / Exam Quotation |
|---|---|---|---|
| **Lala Lajpat Rai ('Punjab Kesari')** | Hisar (Karma-bhoomi 1886-92) | Founded Hisar Arya Samaj (1886) & Hisar Congress (1887) | Practised law at Hisar Court; served on Hisar Municipal Committee; established national protest networks |
| **Pandit Neki Ram Sharma ('Haryana Kesari')** | Kelanga village (Rohtak/Bhiwani) | Weekly Newspaper: 'Sandesh' (संदेश) from Bhiwani | Opposed British recruitment in WWI; famously told Viceroy: 'Mujhe to sampoorna Bharat ki bhoomi chahiye' |
| **Lala Murli Dhar ('Grand Old Man of Punjab')** | Ambala (Born in Palwal) | Founding member of Indian National Congress (1885 Bombay) | Jailed in 1886; renounced title 'Rai Bahadur' & 'Kaisar-i-Hind' medal in protest against Jallianwala Bagh massacre (1919) |
| **Deshbandhu Gupta** | Panipat | Daily Tej (Delhi) & nationalist journalism | Championed integration of Haryana and Delhi in 1932; leading figure in Constituent Assembly |
| **Smt. Kasturba & Mahatma Gandhi Visits** | Palwal & Rohtak | First arrested in Haryana at Palwal Railway Station (10 Apr 1919) | En route to Punjab against Rowlatt Act; Gandhi Smarak established at Palwal railway platform |

# Haryana Legislative Assembly: Composition, Evolution & Reservation

*Key Exam Takeaway: Unicameral house of 90 seats; exactly 17 seats are reserved for Scheduled Castes (SC).*

## ASSEMBLY STRUCTURE & SEAT EVOLUTION

• **Unicameral Architecture:** Single legislative chamber (Haryana Vidhan Sabha) located in the Capitol Complex, Chandigarh (designed by Le Corbusier).

• **Evolution of Assembly Seats:**

• • **At State Formation (1 Nov 1966):** 54 seats (10 reserved for SC).

• • **First General Elections (March 1967):** Increased to 81 seats (15 reserved for SC).

• • **Delimitation of 1977:** Increased to 90 seats (17 reserved for SC); remains constant until 2026 freeze.

• **Proposed New Assembly Building:** 10 acres of land allocated near Railway Station light point in Chandigarh for expanded delimitation capacity (>126 seats).

## 17 RESERVED CONSTITUENCIES (SC) & EXCEPTIONS

• **17 Reserved Constituencies (SC):**

• 1. Kalka/Mulana (Ambala)  2. Sadhaura (Yamunanagar)

• 3. Shahbad (Kurukshetra)  4. Guhla (Kaithal)

• 5. Nilokheri (Karnal)  6. Israna (Panipat)

• 7. Kharkhoda (Sonipat)  8. Narwana (Jind)

• 9. Uklana (Hisar)  10. Bawanikhera (Bhiwani)

• 11. Kalanaur (Rohtak)  12. Jhajjar (Jhajjar)

• 13. Bawal (Rewari)  14. Pataudi (Gurugram)

• 15. Hodal (Palwal)  16. Rania/Kalanwali (Sirsa)

• 17. Ratia (Fatehabad)

• **Districts with ZERO Reserved Seats:** Panchkula, Charkhi Dadri, Mahendragarh, and Nuh (Mewat).

# Parliamentary Representation: 10 Lok Sabha & 5 Rajya Sabha Seats

*Key Exam Takeaway: Haryana sends 10 MPs to Lok Sabha (2 reserved: Ambala & Sirsa) and 5 MPs to Rajya Sabha.*

| Constituency Type | Total Seats | Specific Constituencies | Key Screening & Electoral Details |
|---|---|---|---|
| **Lok Sabha (Lower House)** | 10 Seats | 1. Ambala (SC) 2. Kurukshetra<br>3. Karnal 4. Sonipat<br>5. Rohtak 6. Sirsa (SC)<br>7. Hisar 8. Bhiwani-Mahendragarh<br>9. Gurgaon 10. Faridabad | 2 Reserved Seats for SC:<br>• Ambala (SC)<br>• Sirsa (SC)<br>Gurgaon is largest electorate; Sonipat & Karnal major agricultural hubs. |
| **Rajya Sabha (Upper House)** | 5 Seats | Elected by 90 MLAs of Haryana Vidhan Sabha through proportional representation (STV) | Staggered 6-year tenure; one-third members retire every two years; represents state interest at Center. |
| **Electoral Demographics** | 22 Districts | Over 2.01 Crore registered voters in Haryana electorate | Highest voter population in Badshahpur assembly (Gurugram); lowest in Narnaul. |
| **Electoral Boundaries** | Delimitation 2008 | Merged Mahendragarh with Bhiwani; carved out modern Gurgaon and Faridabad PC | Assembly segments per Lok Sabha constituency: Exactly 9 Assembly segments each (9 x 10 = 90). |

Election Commission of India (ECI); Lok Sabha Secretariat.

# Chief Ministers of Haryana: Historic Tenures & Landmark Milestones

*Key Exam Takeaway: Pt. BD Sharma was 1st CM; Rao Birender Singh was 1st non-Congress CM; Bhajan Lal served longest.*

| Chief Minister | Party / Alignment | Tenure Milestone | Key Historical / Exam Fact |
|---|---|---|---|
| **Pt. Bhagwat Dayal Sharma** | Indian National Congress | 1 Nov 1966 - 23 Mar 1967 | First Chief Minister of Haryana; hailed from Beri (Jhajjar); later Governor of Odisha & MP. |
| **Rao Birender Singh** | Vishal Haryana Party (VHP) | 24 Mar 1967 - 2 Nov 1967 | First non-Congress CM; first elected male Speaker of Vidhan Sabha; founded regional Vishal Haryana Party (Symbol: Rising Sun). |
| **Bansi Lal (3 Non-Consecutive Eras)** | Congress / Haryana Vikas Party | 1968-75, 1985-87, 1996-99 (~4,268 total days) | 'Architect of Modern Haryana'; 100% rural electrification by Nov 1970; lift irrigation; alcohol prohibition (1996-98). |
| **Banarsi Das Gupta** | Congress | 1975-77 & 1990 | Served as Speaker, Deputy CM, and Chief Minister of Haryana. |
| **Chaudhary Devi Lal ('Tau')** | Janata Party / Lok Dal | 1977-79 & 1987-89 | 'King Maker'; twice Deputy Prime Minister of India; introduced Old Age Pension & farm loan waiver. |
| **Bhajan Lal ('Ph.D of Politics')** | Congress / Haryana Janhit Congress | 1979-86 & 1991-96 (Longest cumulative tenure: 3,952 days) | Longest-serving Chief Minister of Haryana; mastered coalition engineering and non-Jat consolidation. |
| **Om Prakash Chautala** | INLD | Sworn in 5 times (1989, 90, 90, 91, 1999-2005) | Highest number of oath ceremonies as CM (5 times); shortest single tenure (6 days in July 1990). |
| **Bhupinder Singh Hooda** | Congress | 5 Mar 2005 - 19 Oct 2014 (Continuous 9 yrs 7 mos) | Longest continuous single stretch as Chief Minister; IMT Manesar and education hubs. |

Haryana Vidhan Sabha Secretariat; General Administration Department (GAD).

# Governors of Haryana & Legacies of Devi Lal and Bansi Lal

*Key Exam Takeaway: Dharma Vira was 1st Governor; B.N. Chakravarty served longest; O.P. Verma served shortest (5 days).*

## GUBERNATORIAL PROFILE & PRESIDENT'S RULE

• **First Governor:** Shri Dharma Vira (1 Nov 1966 - 14 Sep 1967).

• **Longest-Serving Governor:** B.N. Chakravarty (15 Sep 1967 - 26 Mar 1976; 8+ years; died in office; B.N. Chakravarty Canal named after him).

• **Shortest-Serving Governor:** Justice O.P. Verma (2 July 2004 - 7 July 2004; barely 5 days).

• **First Female Speaker:** Smt. Shanno Devi (first female speaker of any state legislature in India!).

• **President's Rule in Haryana (3 Times):**

• 1. 21 Nov 1967 - 21 May 1968 (Rao Birender Singh regime fall)

• 2. 30 Apr 1977 - 21 Jun 1977 (Post-Emergency dissolution)

• 3. 6 Apr 1991 - 23 Jul 1991 (Om Prakash Chautala regime fall)

## DEVI LAL VS BANSI LAL: MODERN ARCHITECTS

• **Chaudhary Devi Lal ('Jan Nayak' / 'Tau'):**

• • Born in Teja Khera (Sirsa); organized historic 'Nyaya Yudh' in 1985-87.

• • Deputy Prime Minister of India (1989-1991) under V.P. Singh & Chandra Shekhar.

• • Introduced pioneering Old Age Pension (वृद्धावस्था पेंशन) of Rs 100/month in 1987 (now Rs 3,000/month).

• • Waived rural farmer and artisan loans up to Rs 10,000.

• **Chaudhary Bansi Lal ('Iron Man of Haryana'):**

• • Born in Golagarh (Bhiwani); modernized state infrastructure.

• • Spearheaded 100% rural electrification achieved on 29 Nov 1970 (felicitated by Indira Gandhi at Ucchani, Karnal).

• • Pioneered tourism naming after birds (Blue Jay, Skylark, Magpie, Parakeet).

# Punjab & Haryana High Court & Panchayati Raj Governance

*Key Exam Takeaway: High Court began in Lahore (1919); Haryana was 2nd state to mandate educational criteria for Panchayats.*

## PUNJAB & HARYANA HIGH COURT

• **Judicial Lineage:** Established at Lahore on 21 March 1919 under Letters Patent issued by King George V.

• **First Indian Chief Justice:** Sir Shadi Lal of Rewari (appointed Chief Justice of Lahore High Court in 1920).

• **Post-Partition Shift:** Shifted to Shimla (Peterhoff) in 1947; relocated to Le Corbusier-designed building in Chandigarh in 1955.

• **First Chief Justice post-1966:** Justice Mehar Singh was Chief Justice when Haryana was carved out on 1 Nov 1966.

• **Sanctioned Strength:** 85 judges (64 permanent, 21 additional); shared jurisdiction over Punjab, Haryana, and UT Chandigarh.

## PANCHAYATI RAJ & EDUCATIONAL CRITERIA MILESTONE

• **Haryana Panchayati Raj Act 1994:** Passed on 22 April 1994 enforcing 73rd Constitutional Amendment (Three-tier: Gram Panchayat, Panchayat Samiti, Zila Parishad).

• **Historic Amendment (2015):** Mandated minimum educational qualification to contest Panchayat polls (10th pass for General males, 8th pass for SC males & women, 5th pass for SC women).

• **Mandatory Sanitation & Defaulter Check:** Must have a functional toilet at home; no power bill arrears; no cooperative loan default.

• **Upheld by Supreme Court:** Validated in landmark case *Rajbala v. State of Haryana (2015)*.

• **Panchayat 2.0 Reforms:** Inter-District Council, Gram Darshan portal, and 50% reservation for women in Panchayati Raj institutions.

# Agrarian Profile: Cropping Seasons, Land Utilization & Production

*Key Exam Takeaway: Haryana contributes ~15% of wheat & ~10% of rice to the National Central Foodgrain Pool.*

## LAND UTILIZATION & GRAIN DOMINANCE

• **Net Sown Area:** ~3.52 million hectares (>80% of total geographical area), reflecting extraordinary land cultivation efficiency (national average ~43%).

• **Cropping Intensity:** Approximately 185% - 190%, driven by intensive tubewell and canal irrigation networks.

• **Total Foodgrain Production:** Surpasses 180-185 lakh metric tonnes annually.

• **National Contribution:** Despite occupying only 1.34% of India's landmass, Haryana is the 2nd largest contributor of wheat and 3rd largest of rice to the Central Pool (PDS / FCI).

• **Gross Irrigated Area:** ~89% of gross cropped area is under assured irrigation (Tubewells ~55%, Canals ~44%).

## THREE CROPPING SEASONS OF HARYANA

• **Rabi Season (Winter / Asadhi - असाढ़ी):**

• • **Sowing:** Oct - Dec; Harvesting: April - May.

• • **Major Crops:** Wheat (Gehun), Mustard (Sarson), Gram (Chana), Barley (Jau), Potato.

• **Kharif Season (Monsoon / Sawani - सावणी):**

• • **Sowing:** June - July; Harvesting: Sep - Nov.

• • **Major Crops:** Paddy (Dhan), Cotton (Narma/Kapas), Pearl Millet (Bajra), Sugarcane, Maize.

• **Zaid Season (Summer / Jayad):**

• • **Sowing:** March - June (between Rabi and Kharif).

• • **Major Crops:** Fodder (Jowar/Barseem), Watermelon, Muskmelon, Cucumber, Seasonal vegetables.

# Crop Belts: Taraori Basmati Rice, Wheat, Mustard, Bajra & Cotton

*Key Exam Takeaway: Karnal is the 'Rice Bowl of Haryana'; Taraori Basmati accounts for over 60% of India's Basmati exports.*

| Crop Category | Leading Districts | Key Agro-Economic Feature | FSO Food Science / Quality Standard |
|---|---|---|---|
| **Basmati Rice (Dhan)** | Karnal, Kaithal, Kurukshetra, Panipat (Rice Bowl) | Taraori Basmati (HBC-19) world famous for aroma & elongation; accounts for >60% of national Basmati exports | Geographical Indication (GI) tagged; 2-acetyl-1-pyrroline (2-AP) compound confers distinctive aroma |
| **Wheat (Gehun)** | Sirsa, Jind, Hisar, Karnal, Fatehabad | Sirsa leads in absolute wheat output; high-yielding varieties WH-1105, HD-2967, HD-3086 dominant | Subject to moisture limit (<12%) and foreign matter inspection under FSSAI cereals regulation |
| **Rapeseed & Mustard (Rai/Sarson)** | Mahendragarh, Bhiwani, Rewari, Charkhi Dadri | Mahendragarh is the premier mustard producing hub; cold-pressed oil extraction in local expellers | Erucic acid regulation (<2.0% in canola types; FSSAI blending limits); pungency due to allyl isothiocyanate |
| **Pearl Millet (Bajra)** | Bhiwani, Mahendragarh, Rewari, Hisar | Staple dryland cereal; highly drought-resilient; Haryana covers Bajra under Bhavantar Bharpayee Yojana | Rich in iron (Fe) and dietary fiber; classified as Nutri-Cereal under International Year of Millets |
| **Cotton (Narma / Kapas)** | Sirsa, Fatehabad, Hisar, Jind ('White Gold') | Sirsa is the undisputed cotton capital of Haryana; Bt-Cotton dominates >95% acreage | Cottonseed oil is refined for edible consumption; rich in tocopherols (Vitamin E) |
| **Sugarcane (Ganna)** | Yamunanagar, Karnal, Ambala, Sonipat | Yamunanagar leads in production; Saraswati Sugar Mills (Yamunanagar) is one of Asia's largest mills | Sucrose recovery percentage (~9.5% - 10.5%); jaggery (Gur) manufacturing cluster at Shahbad |

Agricultural & Processed Food Products Export Development Authority (APEDA); HAFED.

# Murrah Buffalo ('Black Gold') & Indigenous Hariana Cattle

*Key Exam Takeaway: Murrah buffalo is Haryana's 'Black Gold' (Kala Sona); Jind is known as the 'Home of Murrah'.*

## MURRAH BUFFALO ('BLACK GOLD')

• **Origin & Homeland:** Rohtak, Jind, Hisar, and Fatehabad districts; Jind is reverently dubbed 'The Capital / Home of Murrah'.

• **Physical Conformation:** Jet-black color, tightly curled spiral horns (like जलेबी), wedge-shaped body, short legs, thin neck, and deep capacious udder.

• **Exceptional Milk Yield:** 2,000 - 3,500 kg per lactation; champions exceed 25-30 liters/day.

• **High Milk Fat Content:** Average fat 7.0% - 8.5%; preferred for ghee, paneer, and mawa production.

• **Global Sire Export:** Murrah bulls exported globally (Brazil, Italy, Bulgaria, Philippines) to upgrade native water buffalo stocks.

• **Economic Title:** Reverently hailed as 'Black Gold' (काला सोना) and 'The Tractor of Rural Economy'.

## HARIANA CATTLE & GLF HISAR

• **Hariana Cattle Breed (Bos indicus):**

• • Native dual-purpose cattle breed originating in Rohtak, Hisar, Jind, and Gurugram.

• • Bullocks renowned nationwide for unmatched draught power, endurance, and swift road transport.

• • **Coat:** White or light grey; long narrow face, flat forehead with convex center.

• **Sahiwal & Tharparkar Propagation:**

• • Promoted under Rashtriya Gokul Mission; Sahiwal cattle breeding farm at Government Livestock Farm (GLF) Hisar.

• **Government Livestock Farm (GLF), Hisar:**

• • One of the largest livestock breeding farms in Asia; established in 1809 as a camel breeding stud by Major James Lumsdaine.

• **Gau Seva Aayog:** Statutory Haryana Gau Seva Aayog headquartered at Panchkula regulating ~600+ Gaushalas.

# Premier Research Institutes: NDRI, CIRB, LUVAS & NBAGR

*Key Exam Takeaway: NDRI shifted from Bangalore to Karnal in 1955; cloned world-famous buffalo 'Garima' & 'Pratham'.*

| Research Institute | Headquarters | Establishment Year | Pioneering Research Achievements (FSO Must-Know) |
|---|---|---|---|
| **ICAR - National Dairy Research Institute (NDRI)** | Karnal | Est. 1923 (Bangalore)<br>Shifted to Karnal: 1955 | Deemed University; developed Karan-Swiss and Karan-Fries cattle crossbreeds; cloned buffalo 'Pratham' (1990) & 'Garima' (2009) using Hand-Guided Cloning; developed rapid chemical test kits for milk adulteration (detergents, starch, urea) |
| **ICAR - Central Institute for Research on Buffaloes (CIRB)** | Hisar (Sirsa Road) | 1 February 1985 | Premier institute for buffalo genetics, embryo transfer technology (ETT), and cryopreservation of semen; cloned superior Murrah bull 'Hisar-Gaurav' and 'Assamese' buffalo |
| **Lala Lajpat Rai University of Vet & Animal Sciences (LUVAS)** | Hisar | 2010 (Carved from CCSHAU) | State veterinary university; veterinary pathology, microbiology, zoonotic diseases, and animal disease diagnostic laboratories |
| **ICAR - National Bureau of Animal Genetic Resources (NBAGR)** | Karnal | 1984 (Shifted to Karnal 1985) | Nodal national agency for registration and characterization of new indigenous livestock and poultry breeds in India |
| **Central Sheep Breeding Farm (CSBF)** | Hisar | 1969-70 | Premier central farm breeding high-quality Corriedale and Sonadi rams to boost mutton and fine wool production |

# Food Science Hubs: NIFTEM Kundli, Mega Food Parks & HAFED

*Key Exam Takeaway: NIFTEM Kundli (Sonipat) is an Institute of National Importance under Ministry of Food Processing Industries.*

## NIFTEM KUNDLI: ACADEMIC EPICENTER

• **National Institute of Food Technology Entrepreneurship & Management (NIFTEM):**

• • Located at Kundli, Sonipat district (adjacent to Delhi border on NH-44).

• • Established by Ministry of Food Processing Industries (MoFPI); declared an 'Institute of National Importance' by Parliament under NIFTEM Act 2021.

• • **State-of-the-art Pilot Plants:** Dairy, Fruits & Vegetables, Meat & Poultry, Cereal & Extrusion, and Packaging Technology.

• • **Certified Food Testing Laboratories:** NABL-accredited and FSSAI-notified referral laboratory testing pesticide residues, mycotoxins, heavy metals, and adulterants.

• • **Incubation & Entrepreneurship:** Provides testing and pilot-scale scaleup facilities for food processing MSMEs across North India.

## MEGA FOOD PARKS & HAFED COOPERATIVE NETWORK

• **Mega Food Parks & Industrial Hubs:**

• • **Mega Food Park at Rai, Sonipat:** Houses large-scale fruit and vegetable dehydration, cold storage, and tetra-pack lines.

• • **Mega Food Park at Barhi, Sonipat:** Focuses on bakery, confectionery, and ready-to-eat (RTE) processing.

• • **Mega Food Park at Dabwali, Sirsa:** Promoted by private/cooperative sector for kinnow and citrus processing.

• • **HAFED Mega Food Park at IMT Rohtak:** Advanced milling, cold chains, and multi-crop automated packaging.

• **HAFED (Haryana State Co-op Supply & Marketing Fed):**

• • Largest apex cooperative body in Haryana (Est. 1 Nov 1966).

• • Operates automatic basmati rice mills at Taraori, Ding, and Rania; cold-pressed mustard oil mills at Rewari and Narnaul; cattle feed plants at Rohtak and Saktakhera.

Ministry of Food Processing Industries (MoFPI); HAFED Annual Report; NIFTEM Act, 2021.

# Indo-Israel Centres of Excellence: Precision Farming & Processing

*Key Exam Takeaway: Gharaunda (Karnal) houses the Centre of Excellence for Vegetables; Mangiana (Sirsa) for Fruits.*

| Centre of Excellence (CoE) | District Location | Focal Commodity | Advanced Technology Implemented |
|---|---|---|---|
| **1. CoE for Vegetables** | Gharaunda (Karnal) | Tomato, Capsicum, Cucumber, Seedling nursery | Naturally ventilated polyhouses, climate-controlled hi-tech nurseries, soilless media (cocopeat), fertigation drip lines |
| **2. CoE for Fruits** | Mangiana (Sirsa district) | Kinnow, Pomegranate, Guava, Olive | Canopy management, high-density meadow orchards (guava), micro-irrigation, post-harvest fruit waxing and sorting |
| **3. CoE for Sub-Tropical Fruits** | Ladwa (Kurukshetra) | Mango, Pear, Litchi, Plum | Elite mother block preservation, rejuvenation of senile orchards, fruit fly trap management |
| **4. CoE for Integrated Beekeeping** | Ramnagar (Kurukshetra) | Honey & Bee Products (Apiculture) | Scientific queen bee rearing, honey processing and extraction facility, automated wax comb foundation plant |
| **5. CoE for Citrus Fruits** | Bhuna (Fatehabad district) | Sweet Orange (Malta), Lime, Kinnow | Viral-free certified citrus rootstock propagation, drip-fertigation monitoring |
| **6. CoE for Dairy / Protected Cultivation** | Hisar (CCSHAU campus) & Hodal (Palwal) | Dairy automation & Floriculture | Hydroponic automated fodder production, protected cultivation of gerbera and rose cut-flowers |

Mission for Integrated Development of Horticulture (MIDH); Indo-Israel Agricultural Project (IIAP).

# Industrial Powerhouses: Gurugram, Faridabad, Panipat & Yamunanagar

*Key Exam Takeaway: Gurugram produces ~50% of India's passenger cars; Panipat is the 'Textile / Weaver City'.*

| Industrial City / Hub | Key Economic Identity | Flagship Companies / Units | Key Manufacturing Output |
|---|---|---|---|
| **Gurugram (Millennium City)** | Automobile & IT Software Capital of North India | Maruti Suzuki (Manesar plant), Hero MotoCorp, DLF Cyber City, Google, Microsoft (>500 Fortune 500 offices) | Passenger cars, two-wheelers, financial technology, enterprise software, auto ancillary parts |
| **Faridabad (Industrial Hub)** | Oldest & largest heavy manufacturing city in Haryana | Escorts Group, JCB India, Yamaha Motors, Whirlpool, Goodyear Tyres, Havells | Tractors, earthmoving machinery, refrigerators, industrial rubber tyres, electrical switchgears |
| **Panipat (Textile / Weaver City)** | 'City of Weavers' & National Handloom Capital | IOCL Panipat Refinery (largest in South Asia), National Fertilizer Limited (NFL), Shodapur loom units | Military acrylic blankets (supplies 75% to armed forces), recycled yarn (shoddy wool), carpets, petrochemicals |
| **Yamunanagar (Paper / Timber City)** | National Timber Market & Paper Hub (Abdullapur Mandi) | Ballarpur Industries Limited (BILT), Saraswati Sugar Mills, Jagadhri Railway Workshop (Est. 1952) | Writing and printing paper, plywood (supplies 60% of India's plywood), railway wagons, refined sugar |

Department of Industries & Commerce, Haryana; Haryana State Industrial Development Corp (HSIIDC).

# Specialized Manufacturing Clusters: Rewari, Ambala, Bahadurgarh & Kundli

*Key Exam Takeaway: Rewari is the 'Brass City' (Pital Nagari); Ambala manufactures ~35% of India's scientific apparatus.*

## REWARI, AMBALA & BAHADURGARH

• **Rewari ('Brass City' / पीतल नगरी):** Century-old artisanal brass casting and sheet rolling; prominent manufacturer of brass and copper utensils, brass electrical components, and Tilla-jooti (embroidered footwear).

• **Jagadhri (Yamunanagar):** Renowned for stainless steel and brass household utensils; extensive rolling mills.

• **Ambala ('Scientific City' & 'Mixie City'):** Accounts for ~35% - 40% of India's total scientific laboratory equipment (microscopes, glassware, physics apparatus); also famous for domestic kitchen mixer-grinder manufacturing.

• **Bahadurgarh ('Gateway of Haryana', Jhajjar):** Major footwear hub in India (Relaxo, Action, Aero shoes); sanitaryware capital (Hindware / HSIL plants).

## IMT CORRIDORS: MANESAR, BAWAL & KHARKHODA

• **IMT Manesar (Gurugram):** Integrated Model Township spread over 4,500 acres; houses Maruti Suzuki, Denso, Honda Motorcycle, and National Brain Research Centre (NBRC).

• **IMT Bawal (Rewari):** Heavy automotive component and Japanese industrial corridor (JETRO) cluster on NH-48.

• **IMT Kharkhoda (Sonipat):** Massive 3,200-acre upcoming mega-manufacturing zone; new mega-plant of Maruti Suzuki (capacity 10 lakh cars/year) and Suzuki two-wheelers.

• **Kundli & Rai (Sonipat):** Food processing, stainless steel fabrication, and plastics manufacturing.

• **Dharuhera (Rewari):** Major automobile and industrial hub (Hero MotoCorp flagship assembly plant).

# Mineral Resources of Haryana: Mahendragarh Dominance & Distribution

*Key Exam Takeaway: Mahendragarh is the undisputed mineral capital of Haryana, holding deposits of iron, slate & calcite.*

| Mineral Commodity | Primary District / Geological Belt | Key Mining Localities | Industrial Application / Exam Fact |
|---|---|---|---|
| **Iron Ore (Magnetite/Hematite)** | Mahendragarh district | Antri-Beharipur, Dhanota-Dhancholi, Chhapra | Low to medium grade deposits associated with Aravalli quartzite ridges |
| **Slate Stone (स्लेट पत्थर)** | Rewari & Mahendragarh | Kund village (Rewari district) - World famous | Kund slate is exported globally for architectural roofing, paving, and chalkboards |
| **Limestone & Kankar** | Panchkula (Morni), Rohtak, Bhiwani | Malla, Kharag, and Tundapathar (Panchkula) | Feedstock for Surajpur Cement Factory (ACC - historical) and Charkhi Dadri cement plant |
| **Marble & Calcite** | Mahendragarh (Narnaul) | Bayal, Musnota, Mandi, Sirohi | Variegated marble (black, white, green banded) and pure crystalline calcite for filler industries |
| **Quartz & Silica Sand** | Gurugram, Faridabad, Mahendragarh | Aravalli ridges, Badarpur sand pits (Faridabad) | Glass manufacturing and construction concrete casting |
| **China Clay (Kaolin)** | Gurugram district | Sikanderpur and Alwar border | Sanitaryware, ceramic tiles, and porcelain insulators |
| **Flexible Sandstone (कम्पा पत्थर)** | Charkhi Dadri (Kaliana village) | Kaliana hillock | Unique geological curiosity: naturally bending/flexible stone (Itacolumite) found only here in India! |

Department of Mines & Geology, Haryana; Indian Minerals Yearbook (IBM).

# Census 2011 Master Profile: Population, Density, Growth & Literacy

*Key Exam Takeaway: Total population: 2,53,51,462 (2.09% of India); Faridabad is most dense (2,442/sq km).*

| Census Demographic Parameter | Absolute Value / Figure | Highest District (Maximum) | Lowest District (Minimum) |
|---|---|---|---|
| **Total Population** | 2,53,51,462 (2.53 Crore)<br>(18th most populous in India) | Faridabad (18,09,733)<br>followed by Hisar | Panchkula (5,61,293)<br>followed by Charkhi Dadri (est) / Fatehabad |
| **Decadal Growth Rate (2001-2011)** | 19.90% (National: 17.70%) | Gurugram (+73.14%)<br>(Massive urban migration) | Jhajjar (+8.90%)<br>followed by Ambala (+11.23%) |
| **Population Density** | 573 persons / sq km<br>(National: 382/sq km) | Faridabad (2,442 persons/sq km)<br>followed by Gurugram (1,204) | Sirsa (303 persons/sq km)<br>(Large arid area, low settlement) |
| **Overall Literacy Rate** | 75.55% (National: 74.04%)<br>Male: 84.06% \| Female: 65.94% | Gurugram (84.70%)<br>followed by Panchkula (81.88%) | Nuh / Mewat (54.08%)<br>followed by Fatehabad (67.92%) |
| **Male Literacy** | 84.06% | Rewari (91.44%)<br>followed by Mahendragarh (89.72%) | Nuh / Mewat (69.94%)<br>followed by Fatehabad (76.14%) |
| **Female Literacy** | 65.94% | Gurugram (77.98%)<br>followed by Panchkula (75.99%) | Nuh / Mewat (36.60%)<br>(Severe socio-educational deficit) |
| **Urban-Rural Spread** | Rural: 65.12% (1.65 Cr)<br>Urban: 34.88% (88.42 Lakh) | Most Urbanized: Faridabad (79.51%)<br>followed by Gurugram (68.82%) | Least Urbanized: Nuh / Mewat (11.39%)<br>followed by Mahendragarh (14.41%) |

# Sex Ratio Dynamics, Child Sex Ratio & Beti Bachao Beti Padhao

*Key Exam Takeaway: Overall Sex Ratio in 2011 was 879 (lowest in India); PM launched 'BBBP' from Panipat in 2015.*

## CENSUS 2011 SEX RATIO REALITIES

• **2011 Overall Sex Ratio:** 879 females per 1,000 males (lowest among all 28 Indian states in Census 2011; national average was 943).

• **Highest Sex Ratio (2011):** Nuh / Mewat (907 females/1,000 males), followed by Fatehabad (902) and Rewari (898).

• **Lowest Sex Ratio (2011):** Gurugram (854 females/1,000 males) and Sonipat (856), reflecting severe urban selective abortions.

• **Child Sex Ratio (0-6 Years, 2011):** A catastrophic 834 females per 1,000 males (national average 918); Jhajjar (782) and Mahendragarh (775) were India's worst-affected districts.

• **Screening Question:** The district with highest overall sex ratio in 2011 is Nuh (907), while lowest is Gurugram (854).

## BETI BACHAO BETI PADHAO & RECOVERY

• **Historic Launch of 'Beti Bachao, Beti Padhao' (BBBP):**

• • Launched by Prime Minister Narendra Modi on 22 January 2015 from Panipat, Haryana.

• • Joint flagship mission of Ministry of Women & Child Development, Ministry of Health, and Ministry of Education.

• • **Strict Enforcement:** Severe crackdown on illegal sex determination under PC-PNDT Act (1994) with special task forces, covert sting operations, and reward schemes for informants.

• **Aapki Beti Hamari Beti Scheme:** Financial incentive deposited into LIC account on the birth of first girl child in SC/BPL families and second girl child in all families.

• **Civil Registration System (CRS) Resurgence:** Sex ratio at birth (SRB) improved significantly to ~915-920+ in recent years.

# Folk Theatre (Saang): Kishan Lal Bhat, Deep Chand & Dada Lakhmi Chand

*Key Exam Takeaway: Deep Chand Bahman is the 'Kalidasa / Shakespeare of Haryana'; Lakhmi Chand is 'Surya Kavi'.*

| Folk Theatre Master / Legend | Native Village & Period | Artistic Title / Honor | Pioneering Contribution & Famous Compositions |
|---|---|---|---|
| **Kishan Lal Bhat** | Meerut (Active c. 1730 CE) | Father / Originator of Saang format | Formalized open-air musical dramatic performance staged on wooden platform (Takhtaposh) with Sarangi and Nagada accompaniment |
| **Deep Chand Bahman** | Khanda village (Sonipat district)<br>Active WWI era | 'Kalidasa of Haryana'<br>'Shakespeare of Haryana' | Modernized Haryanvi Saang; introduced disciplined orchestra, musical meters (Alha, Chhand); recruited youth during WWI; authored 'Raja Harishchandra', 'Gopichand' |
| **Pandit Lakhmi Chand ('Dada Lakhmi')** | Jatti Kalan village (Sonipat)<br>(1903 - 1948) | 'Surya Kavi' (सूर्य कवि)<br>'Sufi Saint of Haryana' | Unlettered genius; composed over 25 classic Saangs (Nal-Damyanti, Satyavan-Savitri, Seth Taramal, Puranmal) and 500+ philosophical Ragnis; Research Chair in MDU Rohtak |
| **Pandit Mange Ram** | Sisana village (Sonipat district) | Chief Disciple of Lakhmi Chand | Authored acclaimed poetic Saangs: 'Pingla-Bhartari', 'Dushyant-Shakuntala'; mastered classical meter |
| **Baje Bhagat** | Sisana village (Sonipat district) | Renowned Ragni & Bhajan Master | Authored popular rural folk compositions: 'Raja Mordhwaj', 'Heer Ranjha'; moral aphorisms |
| **Ahmad Baksh Thanesari** | Thanesar (Kurukshetra) | Pioneering Epic Adaptor | Translated Valmiki Ramayana into Haryanvi Saang format; authored 'Gopi Chand' and 'Padmavat' |

# Folk Dances of Haryana: Dhamal, Phag, Khoria, Loor & Ratvai

*Key Exam Takeaway: Dhamal dates back to Mahabharata; Khoria is performed when groom departs; Loor means 'girl'.*

| Folk Dance Form | Cultural Occasion / Season | Gender / Performer Type | Distinctive Feature & Screening Trap |
|---|---|---|---|
| **1. Dhamal Dance (धमाल)** | Post-harvest moonlit nights; Spring | Exclusively Men (Ahirwal/Mahendragarh/Jhajjar) | Oldest dance form dating back to Mahabharata era; performed with Daph, Dholak, and wooden Chimta to celebrate good crop |
| **2. Phag Dance (फाग)** | Phalgun month (around Holi festival) | Men and women together (mixed) | Celebratory rhythm; witty banter with colored water sprinkling and rhythmic clapping |
| **3. Khoria Dance (खोरिया)** | Wedding night after Groom's Barat departs | Exclusively Women in domestic courtyard | Parody and joyful drama performed by women waiting for bride's arrival; men strictly prohibited |
| **4. Loor Dance (लूर)** | Holi season in Bangru linguistic belt | Exclusively Girls / Young Women | 'Loor' means girl in Bangru dialect; question-answer lyric format between two facing lines of dancers |
| **5. Chhathi Dance (छठी)** | 6th day after the birth of male child | Women of the household & neighborhood | Devotional thanksgiving to Bemata; boiled gram (Bakli / बाकली) distributed as prasad |
| **6. Gugga Dance (गुग्गा)** | Bhadrapada month (Gugga Navami) | Male devotees / Bhagats | Religious ritual dance; devotees self-flagellate with iron chains (Chhad/छड़) to honor snake deity Gugga Pir |
| **7. Ratvai Dance (रतवाई)** | Monsoon season in Mewat region | Meo men and women | Accompanied by Deru (small drum) and Bansuri; fast swirling circular formations |
| **8. Daph Dance (डफ)** | Spring harvest / Baisakhi celebration | Male dancers playing large circular frame drums | First state folk dance officially represented in Republic Day Parade (1969) under CM Bansi Lal |

# Traditional Attire, Folk Musical Instruments & Iconic Fairs

*Key Exam Takeaway: Surajkund Mela started in 1987; Khandwa is the turban of rural honor; Daaman is the 52-yard skirt.*

## TRADITIONAL ATTIRE & FOLK ORNAMENTS

• **Male Attire:** Khandwa (खंडवा - ceremonial turban symbolizing dignity and respect), Pagri / Safa, Kurta, and Dhoti.

• **Female Attire:** Daaman (दामण - pleated flared skirt traditionally made from 52 yards / बावन गज of heavy fabric), Kurti (fitted blouse), and Chundri (चूंदड़ी - printed veil).

• **Traditional Ornaments:**

• • **Head & Forehead:** Borla, Tika, Shringarpatti.

• • **Nose:** Nath, Koka, Purli.

• • **Neck:** Hansli (solid silver torque), Kathla, Galshree, Haar.

• • **Hands & Waist:** Pahunchi, Chhan-Kangan, Tagdi (silver waist chain).

• • **Feet:** Kari, Ramjhol, Pajeb.

## MUSICAL INSTRUMENTS & LANDMARK FAIRS

• **Folk Musical Instruments:** Ektara, Sarangi (associated with Saang), Been (snake charmer flute), Algoza (twin flutes), Deru, Nagada, Tasha, Khartal, and Chimta.

• **Surajkund International Crafts Mela (Faridabad):**

• • Annual fair held in February since 1987 in Surajkund Aravalli complex.

• • Features designated Theme State and Partner Country to promote rural crafts.

• **International Gita Mahotsav:** Held annually in Kurukshetra (Brahma Sarovar) on Shukla Ekadashi of Margashirsha (Mokshada Ekadashi).

• **Other Iconic Fairs:** Kapal Mochan Mela (Yamunanagar), Sheetla Mata Mela (Gurugram), Falgu Mela (Pharal, Kaithal - Somvati Amavasya), Mango Mela (Pinjore Gardens, July).

# Sports Capital of India: Olympic Medal Factory & Cash Award Policy

*Key Exam Takeaway: Haryana awards India's highest Olympic cash prizes (Gold: 6 Cr, Silver: 4 Cr, Bronze: 2.5 Cr).*

| Sports Discipline | Legendary Champions from Haryana | Native Village / District | Historic Milestone / Olympic Achievement |
|---|---|---|---|
| **Athletics (Javelin Throw)** | Subedar Major Neeraj Chopra (VSM) | Khandra village (Panipat district) | Independent India's 1st Track & Field Olympic Gold (Tokyo 2020: 87.58m); Olympic Silver (Paris 2024: 89.45m); World Champion |
| **Wrestling (Kushti)** | Sakshi Malik, Bajrang Punia, Phogat Sisters, Ravi Dahiya, Aman Sehrawat | Mokhra (Rohtak), Khudan (Jhajjar), Balali (Charkhi Dadri), Nahri (Sonipat) | Sakshi: 1st Indian female wrestler Olympic medal (Rio 2016 Bronze); Dahiya: Tokyo 2020 Silver; Aman: Paris 2024 Bronze |
| **Shooting** | Manu Bhaker | Goria village (Jhajjar district) | First Indian athlete post-independence to win 2 medals at a single Olympic Games (Paris 2024: 10m Air Pistol individual & mixed team) |
| **Boxing** | Vijender Singh, Amit Panghal, Nitu Ghanghas | Kaluwas (Bhiwani - 'Mini Cuba'), Mayna (Rohtak), Dhanana (Bhiwani) | Vijender: India's 1st Olympic Boxing medal (Beijing 2008 Bronze); Bhiwani Boxing Club (BBC) founded by Coach Jagdish Singh |
| **Field Hockey** | Rani Rampal, Savita Punia, Sandeep Singh, Surinder Kaur | Shahbad Markanda (Kurukshetra district) | Shahbad is renowned as the 'Nursery / Cradle of Indian Women's Hockey' & 'Sansarpur of Haryana' |

# District Profile Matrix Part I: Northern & Central Districts

*Key Exam Takeaway: Quick-revision encyclopedic matrix covering administrative hubs from Panchkula to Rohtak.*

| District | Creation Date | Historical / Nickname | Key Screening Points & Monuments |
|---|---|---|---|
| **Panchkula** | 15 Aug 1995 | 'City of Five Canals' (Panch Kula) | Morni Hills (Karoh 1514m); Pinjore Gardens (Yadavindra Gardens built by Fidai Khan); Mansa Devi Temple; HMT Pinjore; VCBC Vulture Centre |
| **Ambala** | 1 Nov 1966 (Original) | 'Scientific City' / 'Mixie City' | Ambala Cantt (Est. 1843); 1857 Uprising outbreak (May 10); birthplace of Sushma Swaraj; Gurdwara Manji Sahib; Muslim shrines |
| **Yamunanagar** | 1 Nov 1989 | 'Paper City' (Abdullapur Mandi) | Hathnikund & Tajewala; Kalesar NP; Saraswati Sugar Mills; Chaneti Stupa; Topra Ashokan pillar site; Jagadhri Railway Workshop |
| **Kurukshetra** | 23 Jan 1973 | 'Dharmakshetra' / Holy City | Brahma Sarovar & Sannihit Sarovar; Jyotisar (Gita sermon); Sheikh Chilli's Tomb; Kurukshetra University (1st Univ in Haryana, 1956); Kalpana Chawla Planetarium |
| **Kaithal** | 1 Nov 1989 | 'Kapishthala' (Hanuman birthplace) | Tomb of Razia Sultan (1240); Falgu Tirth (Pharal); Anjani Teela; Neem Sahib Gurdwara; Balu archaeological site |
| **Karnal** | 1 Nov 1966 (Original) | 'Rice Bowl' / 'City of Daanveer Karna' | NDRI (1955); CSSRI (1969); IIWBR; NBAGR; Karna Lake; Kalpana Chawla birthplace (1962); Liberty Shoes; Taraori Basmati export hub |
| **Panipat** | 1 Nov 1989 (Recreated 1992) | 'City of Weavers' / 'Textile City' | 3 Battles of Panipat (1526, 1556, 1761); Kabuli Bagh Mosque; Kala Amb; Bu Ali Shah Qalandar Dargah; Altaf Hussain Hali tomb; IOCL Refinery |
| **Sonipat** | 22 Dec 1972 | 'Swarnaprastha' / Golden City | NIFTEM Kundli; OP Jindal Global Univ; Sports University Rai (1st Sports Univ); Deenbandhu Chhotu Ram Univ (DCRUST Murthal); Atlas Cycles (historical) |
| **Rohtak** | 1 Nov 1966 (Original) | 'Rohitaka' / Education Hub | MDU Rohtak (Est. 1976); Pt. BD Sharma Post Graduate Institute of Medical Sciences (PGIMS); Tilyar Lake; Garhi Sampla (Chhotu Ram); Rewari-Rohtak eco zone |

Directorate of Land Records, Haryana; District Gazetteers.

# District Profile Matrix Part II: Western & Southern Districts

*Key Exam Takeaway: Quick-revision encyclopedic matrix covering administrative hubs from Hisar to Charkhi Dadri.*

| District | Creation Date | Historical / Nickname | Key Screening Points & Monuments |
|---|---|---|---|
| **Hisar** | 1 Nov 1966 (Original) | 'City of Steel' / Hisar-e-Firoza | Founded by Firoz Shah Tughlaq (1354); Gujri Mahal; CCSHAU (1970); CIRB (1985); LUVAS (2010); Jindal Steel; Rakhigarhi; Agroha Dham; Blue Bird Lake |
| **Fatehabad** | 15 Jul 1997 | 'Pink City of Haryana' | Founded by Firoz Shah (named after son Fateh Khan); Bhirrana (oldest site ~7500 BCE); Banawali & Kunal; Gorakhpur Nuclear Power Plant (2800 MW) |
| **Sirsa** | 1 Sep 1975 | 'Saraswati Nagari' / 'Arnyavanya' | Largest district by area (4,277 sq km); lowest population density (303); Abubshahar Sanctuary (largest); Ottu Barrage on Ghaggar; Tara Baba Kutiya |
| **Jind** | 1 Nov 1966 (Original) | 'Heart of Haryana' / Jayantipuri | Named after Jayanti Devi; 'Home of Murrah Buffalo'; Bir Bara Ban reserve; Dhamtan Sahib Gurdwara; Safidon (ancient Sarpadevi where Janamejaya performed snake sacrifice) |
| **Bhiwani** | 22 Dec 1972 | 'Chhoti Kashi' / 'Mini Cuba' | City of 300+ temples ('Chhoti Kashi'); Bhiwani Boxing Club; Mitathal & Naurangabad; Devsar & Star Monument; Board of School Education Haryana (BSEH) |
| **Charkhi Dadri** | 1 Dec 2016 | 22nd District of Haryana | Carved from Bhiwani; flexible sandstone (कम्पा पत्थर) at Kaliana; famous Phogat sisters (wrestling); 1996 mid-air plane crash site |
| **Mahendragarh** | 1 Nov 1966 (Original) | Mineral Capital of Haryana | Headquarters at Narnaul (only district whose HQ is not in its own name!); Dhosi Hill (652m, Chyavana Rishi); Birbal ka Chhatta; Jal Mahal; Antri-Beharipur iron |
| **Rewari** | 1 Nov 1989 | 'Brass City' / Ahirwal Epicenter | Rao Tula Ram (1857 hero); Hemu's hometown; Heritage Steam Locomotive Shed (1893, Fairy Queen); Kund slate stone; Bawal IMT; first Arya Samaj branch in Haryana (1880) |
| **Gurugram** | 1 Nov 1966 (Original) | 'Millennium City' / Guru Drona seat | Highest per capita income; Maruti Suzuki; Sheetla Mata Temple; Sultanpur NP; National Brain Research Centre (NBRC) Manesar; International Solar Alliance (ISA) HQ |
| **Faridabad** | 2 Aug 1979 | Industrial City (Baba Farid) | Most populous & dense district (2442/sq km); Surajkund crafts mela; Badkhal lake; Raja Nahar Singh Palace (Ballabhgarh); Escorts & JCB |
| **Palwal** | 15 Aug 2008 | 'Apelva' / Cotton City | Panchvati Temple; Mahatma Gandhi first arrested here (10 Apr 1919); Hodal Chameli forest; lowest recorded forest percentage (~0.71%) |
| **Nuh (Mewat)** | 4 Apr 2005 | Historical Mewat / Satyamevpuram | Highest sex ratio (907) & lowest literacy (54.08%); Kotla lake & Sheikh Musa tomb; Ratvai dance; Chhui Mal pond; Country's first Mobile Court (Punhana, 2007) |

# Landmark Initiatives: Parivar Pehchan Patra, MFMB & Bhavantar

*Key Exam Takeaway: Haryana is 1st state to implement Bhavantar Bharpayee Yojana (price deficit compensation).*

| Flagship Scheme / Portal | Launch Year | Core Objective | Screening Exam Question Focus |
|---|---|---|---|
| **Parivar Pehchan Patra (PPP)** | 2020 | 8-digit unique biometric/digital family ID for every Haryana household | Mandatory single identifier for availing all state DBT, pensions (Old Age), राशन कार्ड, and public scholarship schemes |
| **Meri Fasal Mera Byora (MFMB)** | 2019 | Unified digital portal for self-registration of cultivated crop acreage | Facilitates transparent MSP procurement by FCI/HAFED, crop loss compensation under PMFBY, and financial subsidies |
| **Bhavantar Bharpayee Yojana (BBY)** | 1 Jan 2018 | Price Deficit Financing Scheme for horticulture & agricultural crops | First state in India to launch BBY; compensates farmers if wholesale market price drops below designated benchmark price; covers 21+ horticulture crops & Bajra |
| **Mhara Gaon Jagmag Gaon (MGJG)** | 1 July 2015 | 24-hour uninterrupted three-phase electricity to rural Haryana villages | Launched from Dyalpura (Kurukshetra); linked to voluntary smart meter installation and power bill payment; >5,800+ villages covered |
| **Mukhyamantri Antyodaya Parivar Utthan** | 2021 | Targeted upliftment of poorest families earning <Rs 1.80 Lakh/year | Provides customized skill training, subsidized entrepreneurship loans, and self-employment linkages under Antyodaya Gramin Melas |
| **CM Window & Jan Samvad** | 25 Dec 2014 | Public grievance redressal system launched on Good Governance Day | Direct digital tracking of citizen complaints monitored at Chief Minister's Office (CMO) level |

Department of Information Technology, Electronics & Communication (DITECH); CM Window Dashboard.

# Food Safety Administration in Haryana: Infrastructure & Enforcement

*Key Exam Takeaway: Food Safety Department functions under Food and Drugs Administration (FDA) Haryana, Panchkula.*

## STATUTORY ENFORCEMENT STRUCTURE

• **Apex Regulatory Body:** Food and Drugs Administration (FDA), Haryana (Headquartered at Sector-14, Panchkula).

• **Administrative Hierarchy:**

• • Commissioner of Food Safety (State Secretary / DG Health level).

• • Designated Officers (DO) at District Level (Administrative jurisdiction over licensing and prosecution).

• • Food Safety Officers (FSO) at Field Level (Inspection, sampling, seizures, and compounding).

• **Food Testing Infrastructure:**

• • **State Food Testing Laboratory, Karnal:** Premier NABL-accredited facility testing adulteration, pesticides, and microbial pathogens.

• • State Drug & Food Testing Lab, Chandigarh / Panchkula.

• • **Referral Testing:** Central Food Laboratory (CFL) Kolkata and NIFTEM Kundli.

## MOBILE TESTING & EAT RIGHT INITIATIVES

• **Mobile Food Testing Labs ('Food Safety on Wheels'):**

• • Fleet of customized mobile analytical vans operating across rural mandis, school mid-day meals, and dairy hubs.

• • Conducts instant on-spot testing for milk fat/SNF, synthetic milk (detergent, urea), starch in paneer, and adulterated edible oils.

• **Eat Right India Campaigns in Haryana:**

• • Eat Right Mela and Walkathons organized at Kurukshetra, Panchkula, and Gurugram.

• • **Clean Street Food Hubs:** Certified hygienic street food markets established at Sector-14 Panchkula and Gurugram.

• **Fortification Mandate:**

• • Fortified Wheat Flour (Atta fortified with Iron, Folic Acid, B12) distributed through Public Distribution System (PDS) across all 22 districts to combat stunting and anemia.

# Top 30 High-Frequency Exam Traps: Rapid-Fire Screening Checklist

*Key Exam Takeaway: Must-memorize high-yield factual checkpoints most frequently tested in HPSC & HSSC exams.*

## RAPID-FIRE CHECKPOINTS: 1 TO 15

- **1. Cherrapunji of Haryana:** Chhachhrauli (Yamunanagar).
- **2. Highest Elevation Peak:** Karoh Peak in Morni Hills (1,514 metres).
- **3. Highest Aravalli Peak:** Dhosi Hill in Mahendragarh (652 metres).
- **4. Oldest Harappan Settlement:** Bhirrana (Fatehabad, ~7500 BCE).
- **5. Largest Harappan Settlement:** Rakhigarhi (Hisar, ~550 hectares).
- **6. Clay Plough Excavated at:** Banawali (Fatehabad, by R.S. Bisht).
- **7. Royal Silver Crowns Excavated at:** Kunal (Fatehabad).
- **8. Earliest Garlic Evidence:** Balu (Kaithal).
- **9. Topra Ashokan Pillar Moved by:** Sultan Firoz Shah Tughlaq in 1356 CE.
- **10. 1857 Uprising Started at:** Ambala Cantt (10 May 1857, 9 AM).
- **11. Shaheedi Diwas Observed on:** 23 September (death of Rao Tula Ram).
- **12. First Chief Minister:** Pt. Bhagwat Dayal Sharma (Beri, Jhajjar).
- **13. First Non-Congress CM:** Rao Birender Singh (Vishal Haryana Party).
- **14. First Female Assembly Speaker:** Shanno Devi (1966).
- **15. First Indian Chief Justice:** Sir Shadi Lal (Lahore High Court, 1920).

## RAPID-FIRE CHECKPOINTS: 16 TO 30

- **16. 100% Rural Electrification Achieved on:** 29 Nov 1970 (under Bansi Lal).
- **17. Old Age Pension Introduced by:** Chaudhary Devi Lal in 1987 (Rs 100).
- **18. District with Highest Forest %:** Panchkula (43.87%).
- **19. District with Lowest Forest %:** Palwal (~0.71%) and Fatehabad.
- **20. Largest Wildlife Sanctuary:** Abubshahar (Sirsa, 11,530 ha).
- **21. Smallest Wildlife Sanctuary:** Chhilchhila (Kurukshetra, 28 ha).
- **22. Two Ramsar Wetland Sites:** Sultanpur NP (Gurugram) & Bhindawas (Jhajjar).
- **23. Vulture Breeding Centre Located at:** Bir Shikargah, Pinjore (Panchkula).
- **24. Home / Capital of Murrah Buffalo:** Jind district.
- **25. National Institute of Food Tech (NIFTEM) at:** Kundli (Sonipat).
- **26. Centre of Excellence for Vegetables:** Gharaunda (Karnal).
- **27. Flexible Sandstone (कम्पा पत्थर) at:** Kaliana (Charkhi Dadri).
- **28. Slate Stone Quarry Hub:** Kund village (Rewari).
- **29. Only District with HQ Outside Name:** Mahendragarh (HQ at Narnaul).
- **30. Beti Bachao Beti Padhao Launched on:** 22 Jan 2015 from Panipat.